Managerial Decision Making
Business 38802 (AXP 25)
Autumn 2025

George Wu

Class 1
Introduction to Decision Making

The Team

- George Wu
  - wu@chicagobooth.edu
  - Generally around until 6:30 PM

- Nick Owsley
  - nowsley@chicagobooth.edu
  - https://www.chicagobooth.edu/research/roman/who-we-are/phd-students/nicholas-owsley

How to keep your professor happy…

Canvas

BUSN 38802 88 (Autumn 2024) Managerial Decision Making

Assign To | Edit | ⋮

38802 - Autumn 2024
Managerial Decision Making - Professor George Wu

Administrative Details

- Professor George Wu | wu@ChicagoBooth.edu | Harper Center 426 | +1 (773) 834-0519
- Teaching Assistants: Nick Owsley and Tong Su 38802ta@gmail.com
- Course Times: 2:00-5:00 PM, November 14-16, December 5-6
- Review Section: Thursday, December 5, 5:00-6:00
- Syllabus: **38802 syllabus** .
- Course Content: Daily course content can be accessed in the grid below. Links to handouts and extra resources are found below.

Daily Course Material

- Class 1: Introduction to Decision Making (Thursday, November 14)
- Class 2: Judgment under Uncertainty: Heuristics and Biases (Friday, November 15)
- Class 3 Judgment under Uncertainty: Why Learning is Difficult (Saturday, November 16)
- Class 4: Risky Decision Making (Thursday, December 5)
- Class 5: Choice Architecture/Improving Decision Processes (Friday, December 6)

Class Handouts (will appear after class)

- Class Handouts

Class Resources (will appear after class)

- Extra readings and discussion

Today's Objectives

- Goals: Introduction to Decision Making
  - Decision making process
  - Three different perspectives on decision making: Normative, Descriptive, and Prescriptive

- Outline
  - First half: lecture sprinkled with examples
  - Course outline, requirements, and expectations
  - Second half: John Brown

We’re building on Managerial Psychology

- Emphasize basic (but important) psychological principles
  - Underlying frameworks and structures
  - Build bridges from theory to practice
  - Some will be review from Managerial Psychology
- But…
  - Lots of new psychology
  - The emphasis will be on improving decision making

and … Negotiation

- Effective negotiation requires good decision making and appropriate judgments

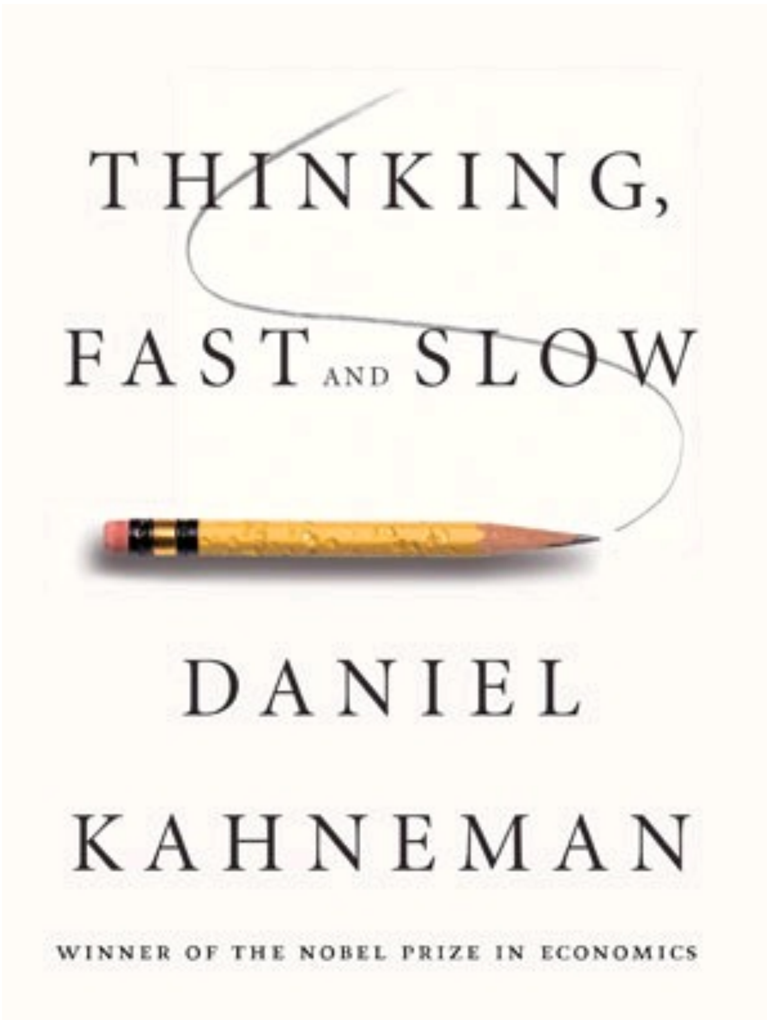
Since 1977…we've been pioneers!

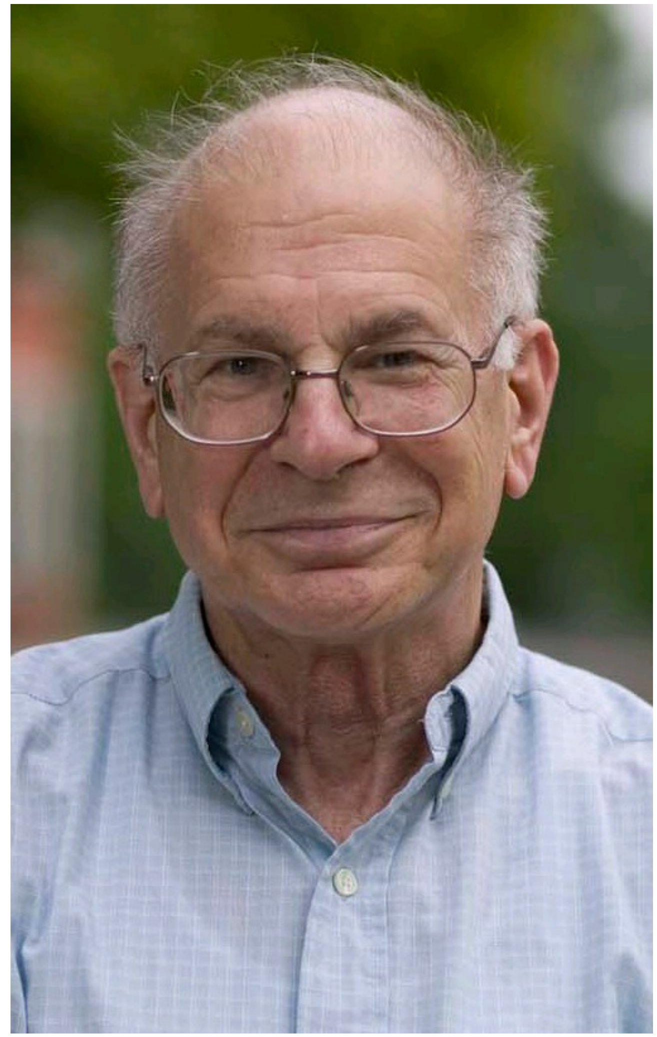
- University of Chicago's Center for Decision Research (now Roman Family CDR) was founded

Hillel J. Einhorn

Robin M. Hogarth

Today…it's mainstream





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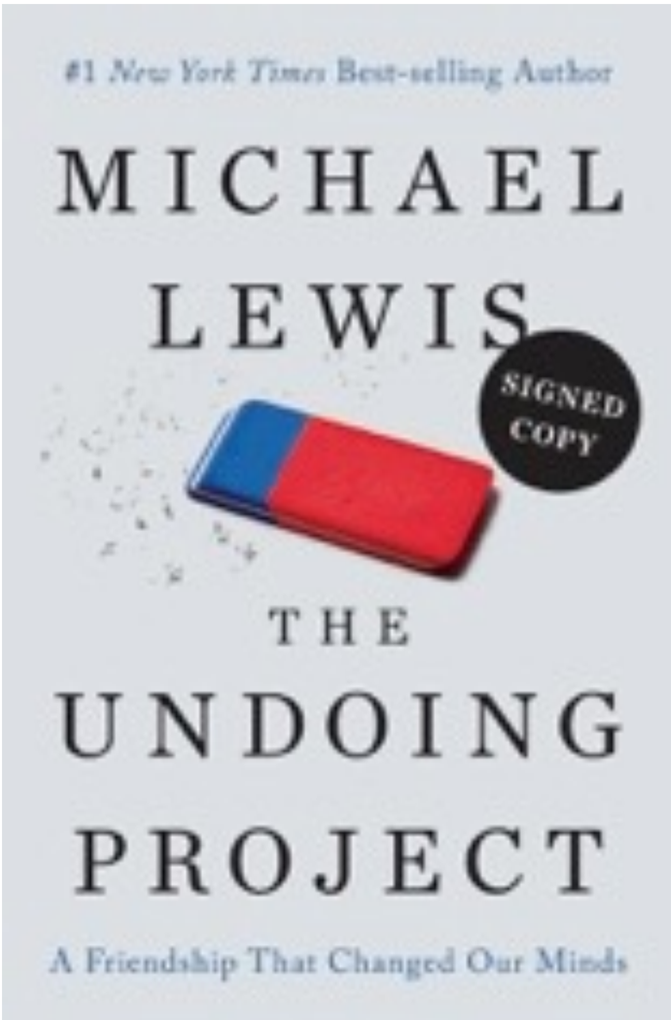

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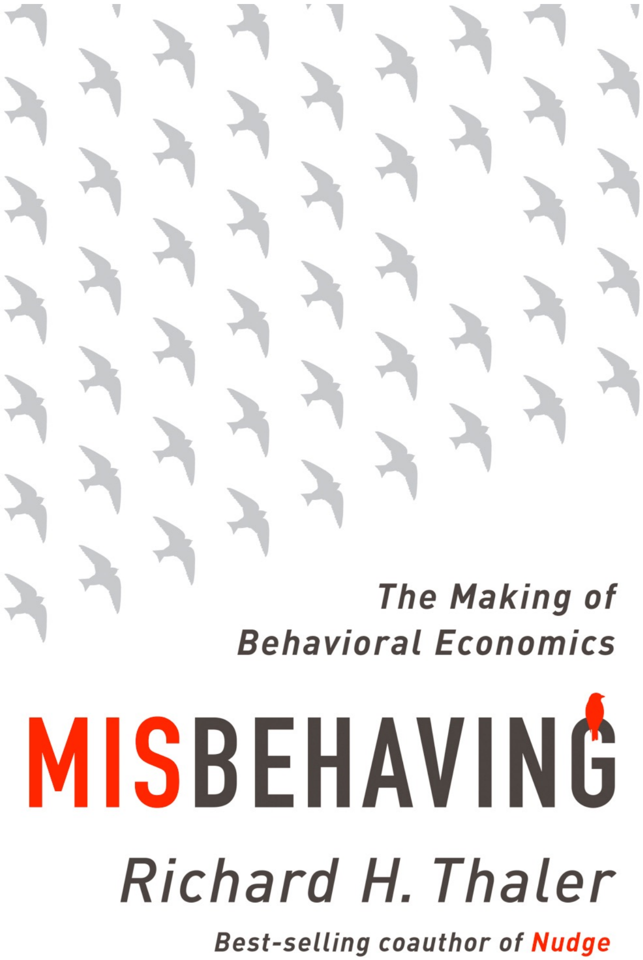
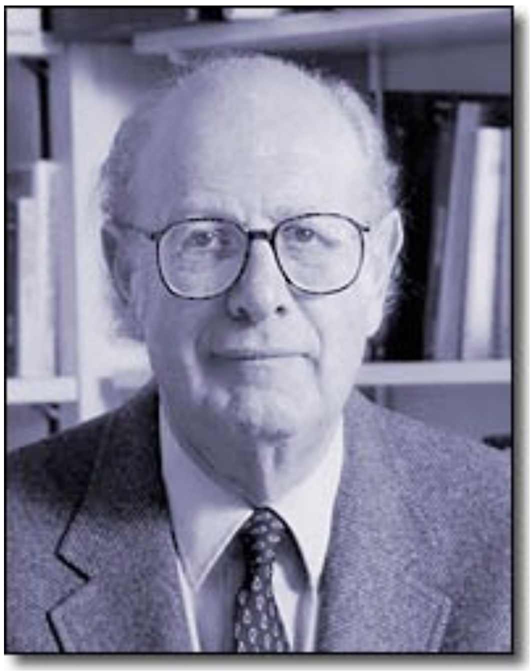


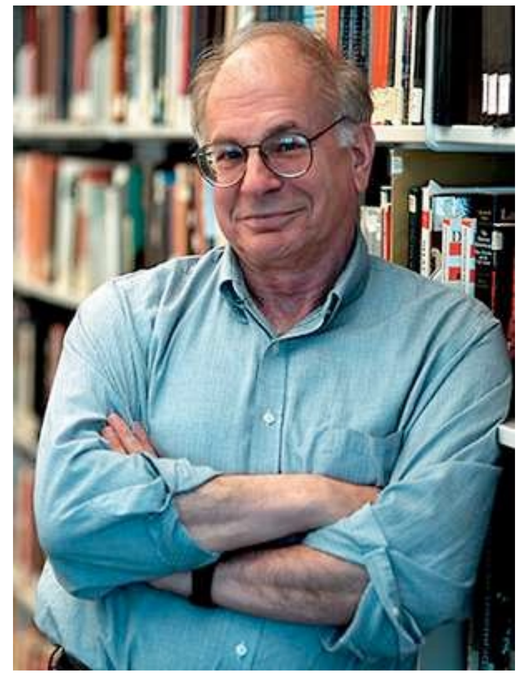
Today…it's mainstream

My Intellectual Journey

Decision Analysis
(Howard Raiffa)

Judgment of Decision Making
(Amos Tversky/Daniel Kahneman)

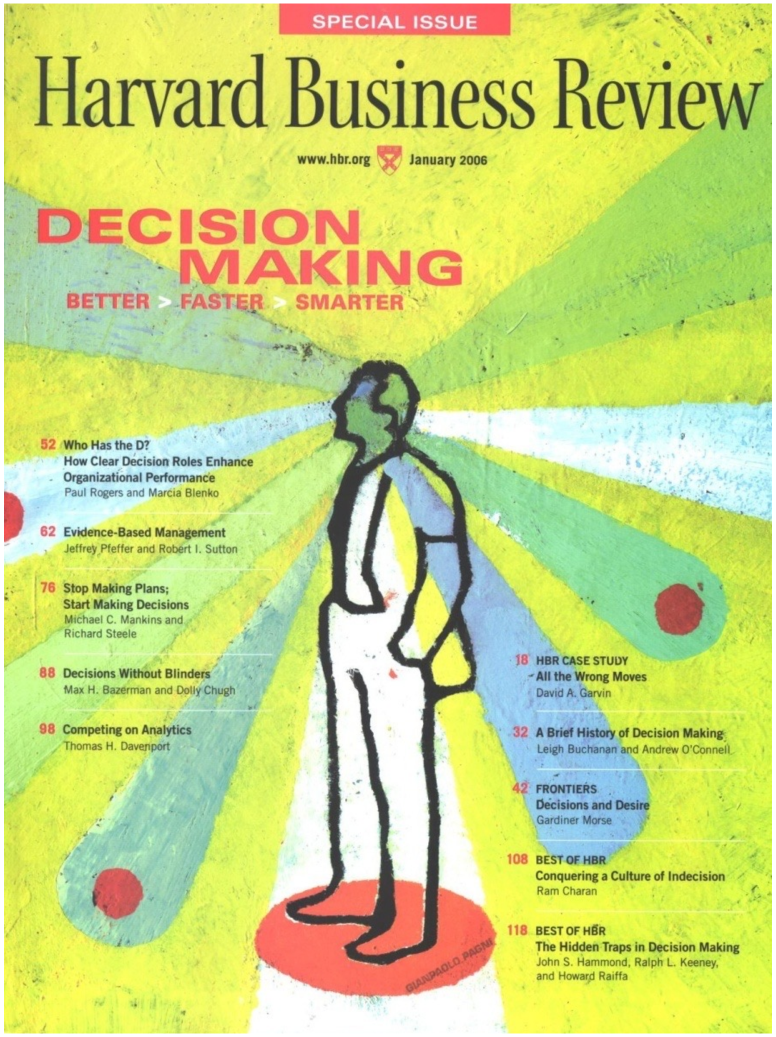




Analytics + Psychology

➔ Great Decision Making

But … why decision making?



Decisions are the essence of management. They're what managers do—sit around all day making (or avoiding) decisions. Managers are judged on the outcomes, and most of them—most of us—have only the foggiest idea how we do what we do.

Thomas Stewart
Editor, *Harvard Business Review*

Decision Making: Two Questions

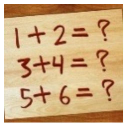
- Why is decision making hard?

- What constitutes a good decision?

Why is decision making difficult?

What constitutes a good decision?

Some Notation



"Do the math"

- Fill in the blanks to make sure that you understand the logic

"Make the connection"

- Tie an idea or concept to something in your life

"Stop and reflect"

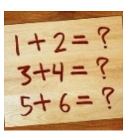
- Step back and think about how you (your organization) does something

Decision Making: Good Process

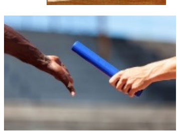
- What is a decision?
  - A <u>costly</u> commitment to a course of action

- Outcomes versus Process

		Outcomes	
		Good	**Bad**
Process	**Good**	√√√	√
	Bad	√	√√√



Why does rewarding outcomes lead to risk avoidance?



Identify examples: Good process/bad outcome + bad process/good outcome

Components of a Good Decision

- I am addressing the appropriate problem
  - I haven't taken the problem as given!
  - I have turned a "problem" into an opportunity
  - I am devoting an appropriate amount of resources
    - STOP Three recent decisions: appropriate amount of resources?

- I have considered my ABCs
  - **A**lternatives
  - **B**eliefs
  - **C**onsequences

- I have avoided major decision traps

Decision Making Components: The ABCs

- <u>A</u>lternatives
  - Identify and articulate (might involve *strategies*)
    - STOP Last big decision: how much time did you spend generating alternatives?
  - Construct and refine (i.e., these are *insight*-driven)
- <u>B</u>eliefs
  - Identify *critical* uncertainties
  - Examine/Collect Information
    - STOP How do *you* determine <u>what</u> information to examine/collect?
  - Quantify
- <u>C</u>onsequences
  - Identify consequences (and objectives addressed by consequences)
  - “Quantify” tradeoffs among objectives

Three Perspectives on Decision Making

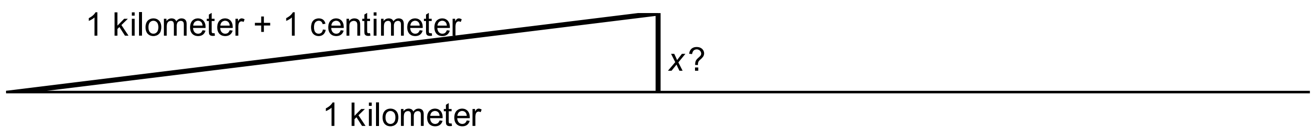
- Normative
  - How <u>should</u> people make decisions?
  - <u>Related concepts</u>: most of your MBA at Booth; rational; optimizing; forward-looking; System 2?
- Descriptive
  - How <u>do</u> people make decisions?
  - <u>Related concepts</u>: *boundedly* rational; emotional; limited cognitive capacity; heuristic; myopic; System 1 (also 2)
- Prescriptive
  - How can you <u>help</u> people (you and others) make better decisions?
  - <u>Related concepts</u>: debiasing; repairs; corrective measures; preventative measures; nudges

Example 1

- Story from the early 1960's:
  - A Columbia University Professor gets a job offer at University of Chicago. Naturally, he likes Chicago more than Columbia. Just when he is ready to accept the Chicago offer, he gets an offer from Berkeley. The choice between Chicago and Berkeley is difficult and causes the professor and his family a lot of anguish.
- Epilogue
  - He decides to stay at Columbia.
- Normative rule

Example 2 (Geometry)

- **Problem**
  - Imagine two one-kilometer long pieces of railroad track, put end to end, and attached to the ground at the extremes. When it gets hot, the track expands by one centimeter, forcing it to rise above the ground. How high is the track off the ground at its peak?



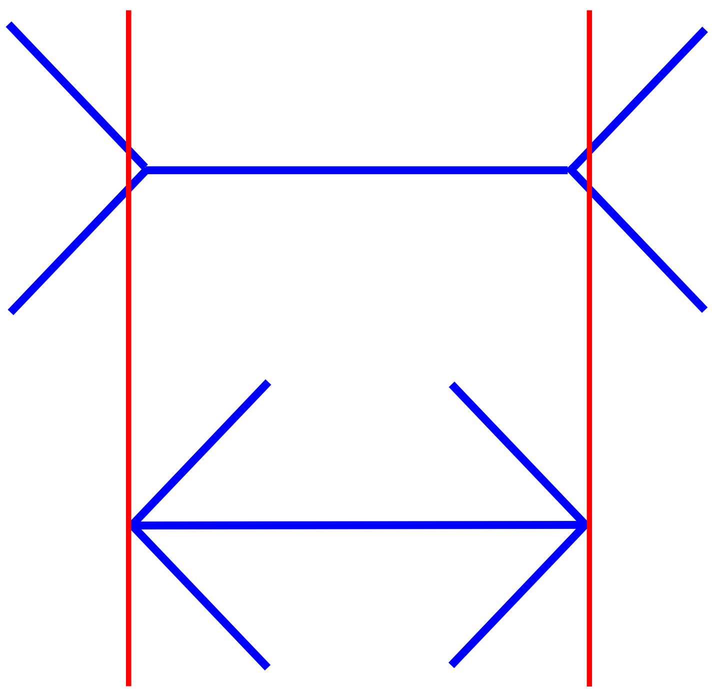
- **Normative rule:**
  - ~~Pythagorean Theorem~~
  - Your best guess? (No calculators!)

Example 2* (Geometry)

- **Problem**
  - A string is attached to the surface (water or land) all along the Equator of the Earth. A second string is elevated above the first string, everywhere exactly one centimeter higher. How much longer is the second string than the first string?

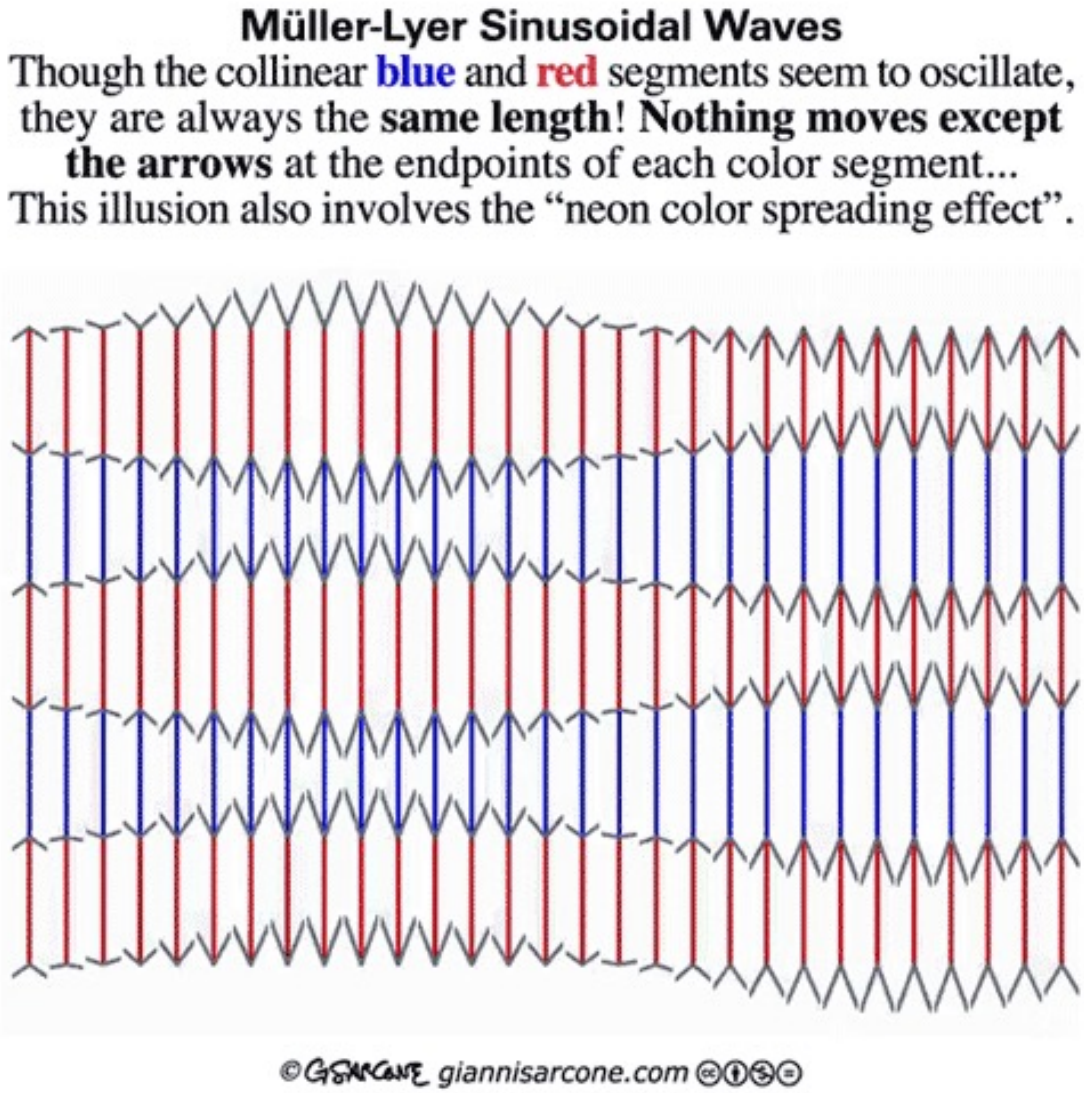
Example 3 (Müller-Lyer)

Which line looks longer?

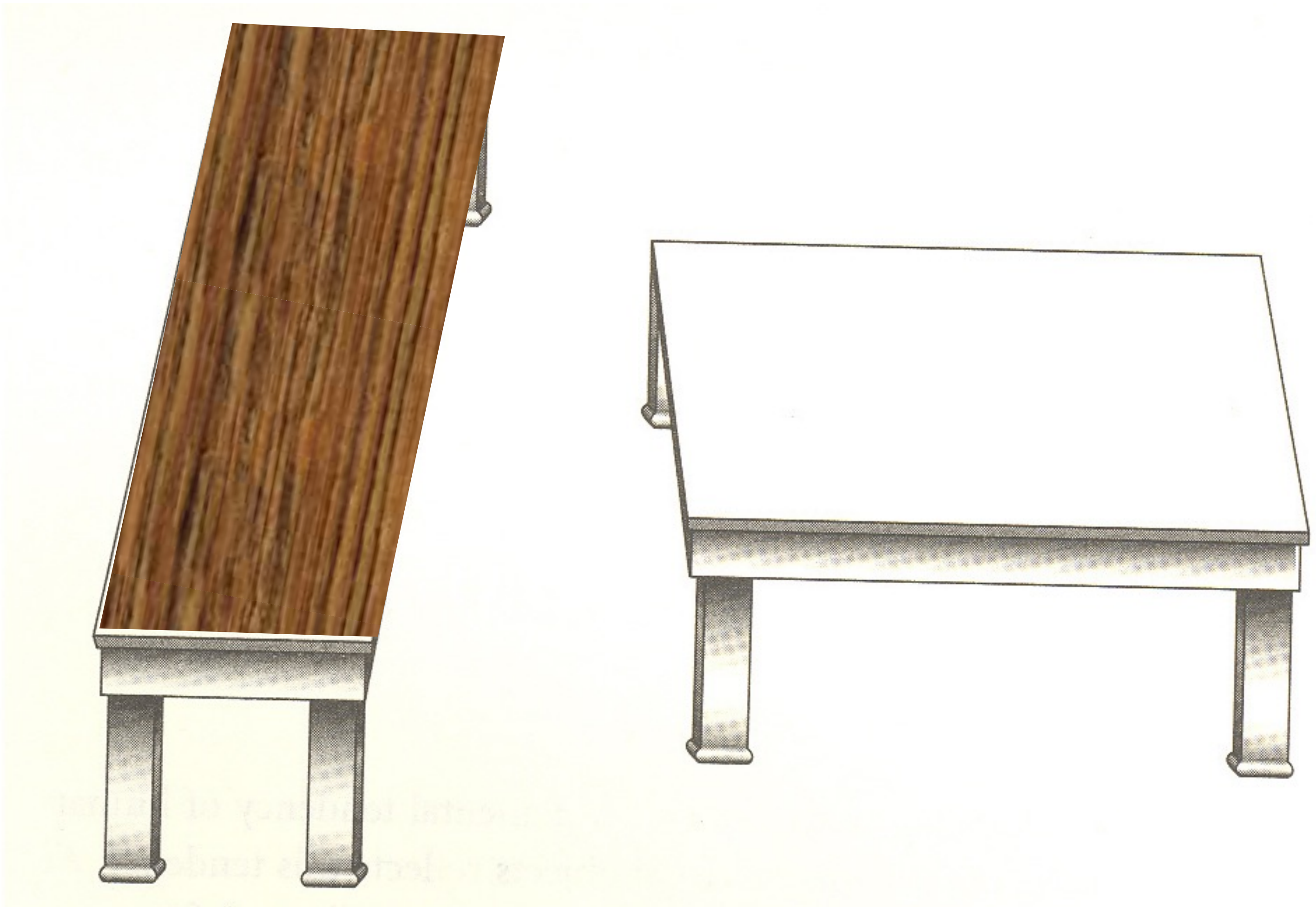


- Normative rule:
  - A line that is longer should be judged to be longer.
- Descriptive reality:
  - Sometimes our vision tricks us.

Example 3 (Müller-Lyer – Even Better)



Example 3* (Another Illusion)



Example 4 (Framing of Decisions)

Problem 1

Assume yourself richer by $<u>300</u> than you are today. You have to choose between a

A. a sure gain of $100
<u>72%</u>

B. a 50% chance to gain $200
a 50% chance to gain nothing
<u>28%</u>

Problem 2

Assume yourself richer by <u>$500</u> than you are today. You have to choose between a

A. a sure loss of $100
<u>36%</u>

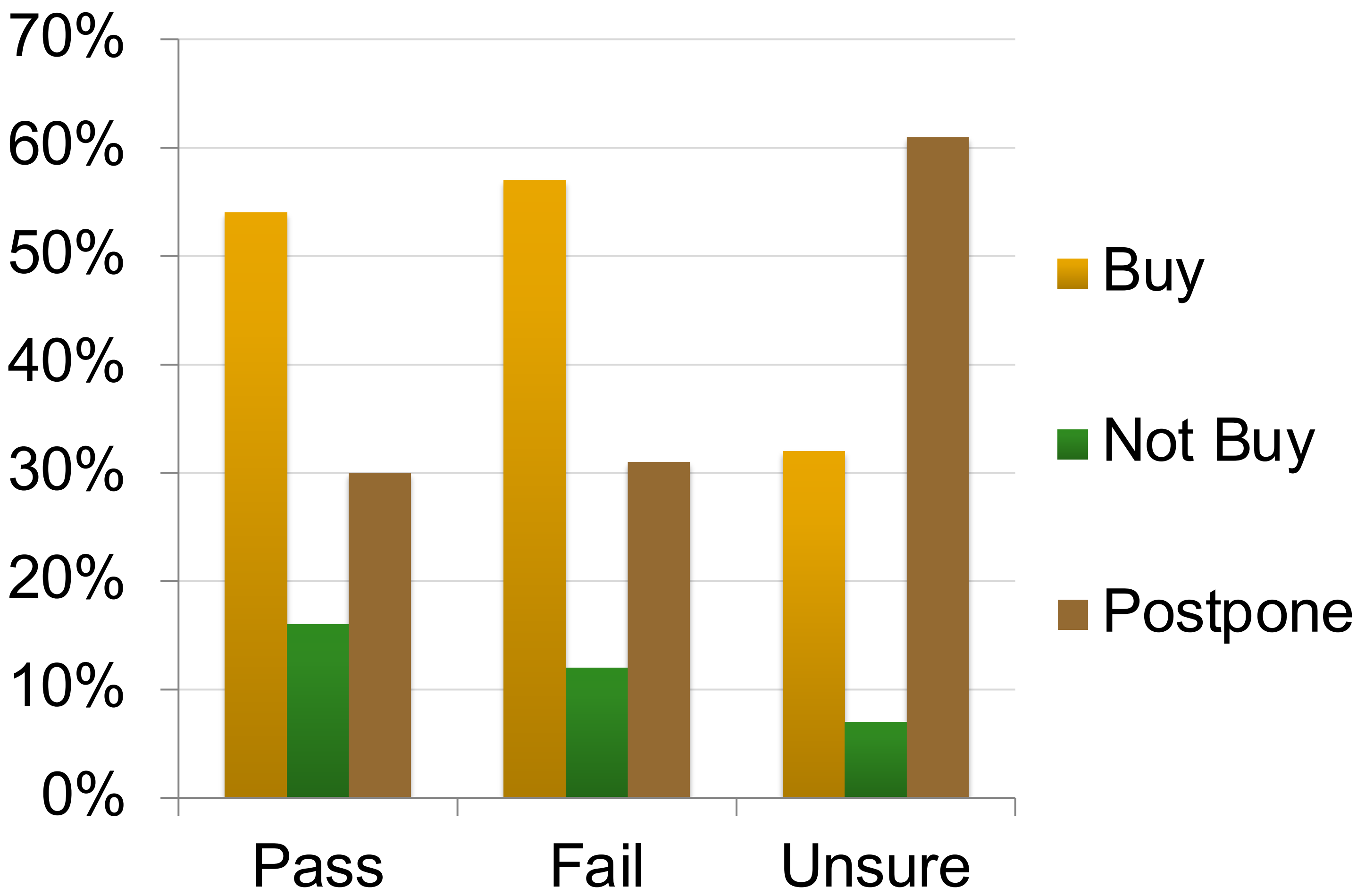
B. a 50% chance to lose nothing
a 50% chance to lose $200
<u>64%</u>

Example 5 (Choosing a Vacation)

- You are a college student who has just taken a tough qualifying exam.
- Three hypothetical scenarios:
  - You passed the exam
  - You failed the exam
  - You are not sure if you have passed or failed

- Decision:
  - Buy an attractively-priced vacation
  - Postpone the choice until tomorrow for $5

Example 5 (Choosing a Vacation)



Example 6 (Monty Hall Problem)

Ask Marilyn™

BY MARILYN VOS SAVANT

You are in error—and you have ignored good counsel—but Albert Einstein earned a dearer place in the hearts of the people after he admitted his errors.

—Frank Rose, Ph.D., University of Michigan

I have been a faithful reader of your column and have not, until now, had any reason to doubt you. However, in this matter, in which I do have expertise, your answer is clearly at odds with the truth.

—James Rauff, Ph.D., Millikin University

May I suggest that you obtain and refer to a standard textbook on probability before you try to answer a question of this type again?

—Charles Reid, Ph.D., University of Florida

Your logic is in error, and I am sure you will receive many letters on this topic from high school and college students. Perhaps you should keep a few addresses for help with future columns.

—W. Robert Smith, Ph.D., Georgia State University

Maybe women look at math problems differently than men.

—Don Edwards, Sunriver, Ore.

You are the goat!

—Glenn Calkins, Western State College

You're wrong, but look at the positive side. If all those Ph.D.s were wrong, the country would be in very serious trouble.

—Everett Harman, Ph.D., U.S. Army Research Institute

Gasp! If this controversy continues, even the postman won't be able to fit into the mailroom. I'm receiving thousands of letters, nearly all insisting that I'm wrong, including one from the deputy director of the Center for Defense Information and another from a research mathematical statistician from the National Institutes of Health! Of the letters from the general public, 92% are against my answer; and of the letters from universities, 65% are against my answer. Overall, nine out of 10 readers completely disagree with my reply.

But math answers aren't determined by votes. For those readers new to all this, here's the original question and answer in full, to which the first readers responded:

"Suppose you're on a game show, and you're given a choice of three doors. Behind one door is a car;

So many readers wrote to say they thought there was *no* advantage to switching (and that the chances became equal) that we published a second explanatory column, affirming the correctness of the original reply and using a shell game and a probability grid as illustrations.

Now we're receiving far *more* mail, and even newspaper columnists are joining in the fray. The day after the second column appeared, lights started flashing here at the magazine. Telephone calls poured into the switchboard, fax machines churned out copy, and the mailroom began to sink under its own weight. Incredulous at the response, we read wild accusations of intellectual irresponsibility and, as the days went by, we were even more incredulous to read embarrassed retractions from some of those same people!

The reaction is understandable. When reality clashes so violently with intuition, people are shaken.

But understanding is strength, so let's look at it again, remembering that the original answer defines certain conditions—the most significant of which is that *the host will always open a losing door on purpose*. (There's no way he can always open a losing door by chance!) Anything else is a different question.

The original answer is still

shows you No. 3; and if the prize is behind No. 3, the host shows you No. 2. So when you switch, you win if the prize is behind No. 2 *or* No. 3. *YOU WIN EITHER WAY!* But if you don't switch, you win only if the prize is behind door No. 1.

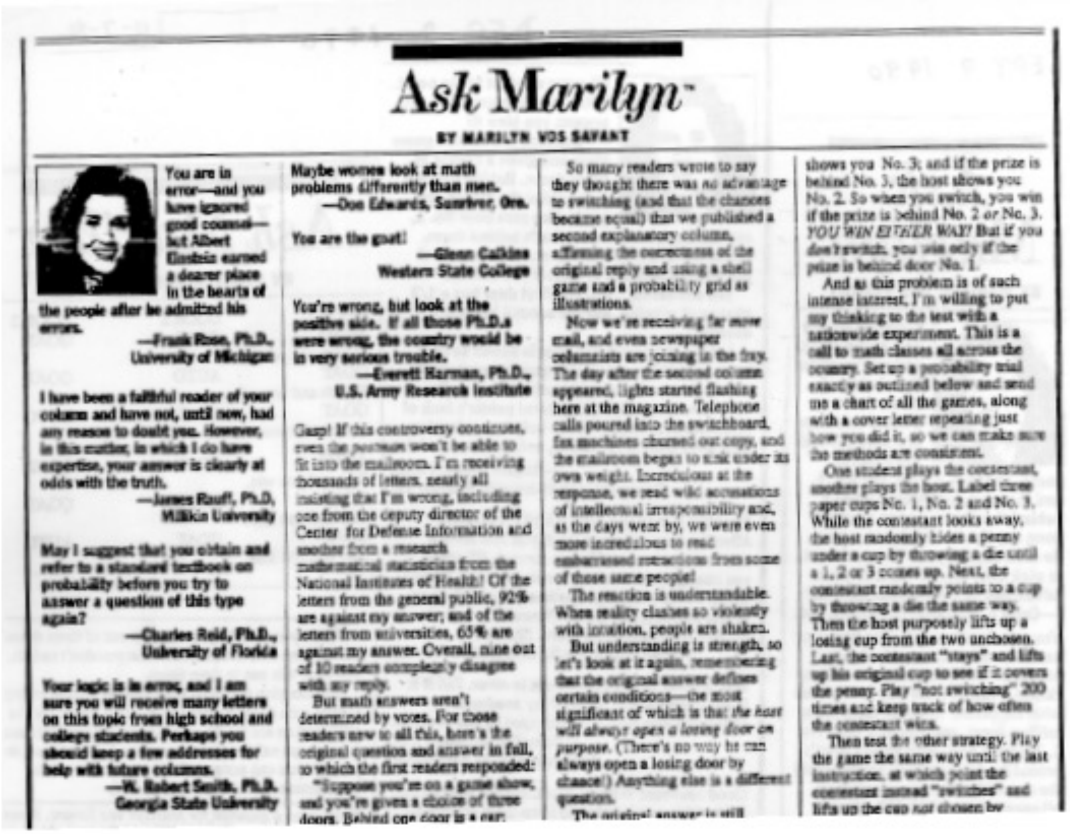
And as this problem is of such intense interest, I'm willing to put my thinking to the test with a nationwide experiment. This is a call to math classes all across the country. Set up a probability trial exactly as outlined below and send me a chart of all the games, along with a cover letter repeating just how you did it, so we can make sure the methods are consistent.

One student plays the contestant, another plays the host. Label three paper cups No. 1, No. 2 and No. 3. While the contestant looks away, the host randomly hides a penny under a cup by throwing a die until a 1, 2 or 3 comes up. Next, the contestant randomly points to a cup by throwing a die the same way. Then the host purposely lifts up a losing cup from the two unchosen. Last, the contestant "stays" and lifts up his original cup to see if it covers the penny. Play "not switching" 200 times and keep track of how often the contestant wins.

Then test the other strategy. Play the game the same way until the last instruction, at which point the contestant instead "switches" and lifts up the cup *not* chosen by


LET'S
MAKE
A
DEAL

Example 6 (Monty Hall Problem)



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"Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?" (*Parade Magazine,* September 9, 1990)

Monty Hall Problem: Assumptions

- Additional Assumptions
  - The host knows which door hides the car.
  - The host will never reveal the car.
  - If the host has a choice of doors to open (i.e., the car is behind the door you have chosen), he will choose randomly.

WORLD

This year's must-have gift: A goat

Thursday, December 23, 2004 Posted: 2:08 PM EST (1908 GMT)

Goats provide milk, offspring, and manure for crops.

LONDON, England (CNN) -- Forget the iPod -- goats and chickens are the must-have Christmas presents this year.

Shoppers have already snapped up more than 31,000 goats and 51,000 broods of chickens under the Oxfam Unwrapped scheme, which sends animals to people in areas of famine around the world.

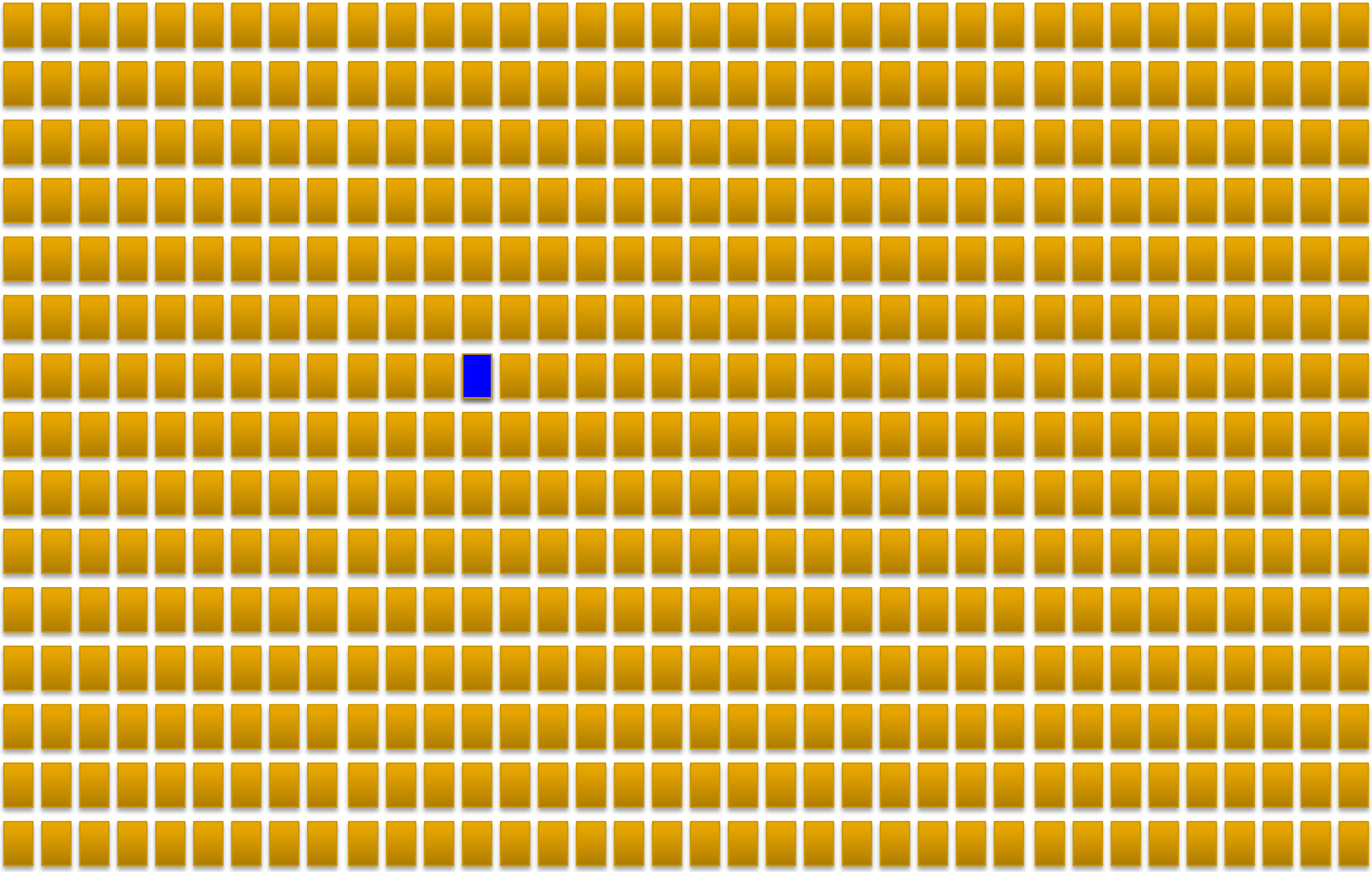
And the Catholic Agency for Overseas Development (Cafod) has sold 13,000 goats at £25 (U.S.$48) each.

Monty Hall Problem: Assumptions

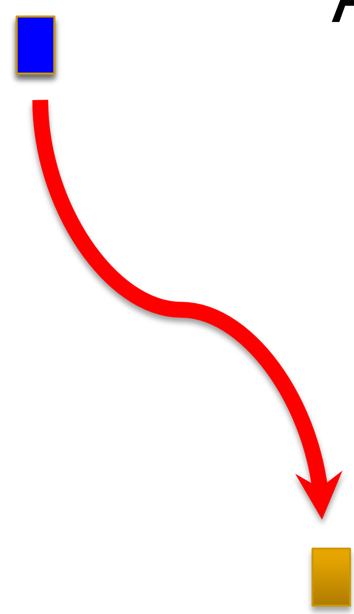
- Additional Assumptions
  - The host knows which door hides the car.
  - The host will never reveal the car.
  - If the host has a choice of doors to open (i.e., the car is behind the door you have chosen), he will choose randomly.
  - You prefer a car to a goat.

What would you do?

540 doors…



540 doors…



Switching:
A) 1-in-2 chance (50%)
or
B) 539-in-540 chance (99.8%)

Monty Hall Problem: Normative

Suppose you have chosen door 1 (thus, initially there is a 1-in-3 chance that the car is behind this door).

- Case 1 (The car behind door 1):

	Stick	Switch
Monty Hall opens door 2	W	L
Monty Hall opens door 3	W	L

- Case 2 (The car behind door 2):

	Stick	Switch
Monty Hall opens door 3	L	W

- Case 3 (The car behind door 3):

	Stick	Switch
Monty Hall opens door 2	L	W

- Summary**:** Your chances are 2/3 if you switch

	Stick	Switch
Behind door 1	W	L
Behind door 2	L	W
Behind door 3	L	W

Monty Hall Problem: Descriptive

Most people (including some mathematicians) think it doesn't matter

- "You blew it, and you blew it big! I'll explain: After the host reveals a goat, you now have one-in-two chance of being correct. Whether you change your answer or not, the odds are the same...Shame!"

Scott Smith, Ph.D., University of Florida
(*Parade*, December 2, 1990)

Monty Hall Simulation

Trial	Prize	Your initial door	door revealed	Stick	Switch
1					
2	3		[illegible]	0	1
3	3	1	2	0	1
4	2	1	3	0	1
5	3	1	2	0	1
6	1	1	2	1	0
7	2	1	3	0	1
8	2	1	3	0	1
9	2	1	3	0	1
10	2	1	3	0	1
11	3	1	2	0	1
12	1	1	3	1	0
13	1	1	3	1	0
14	1	1	2	1	0
15	2	1	3	0	1
16	3	1	2	0	1
17	3	1	2	0	1
18	2	1	3	0	1
19	1	1	2	1	0
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24	2	1	3	0	1
25	3	1	2	0	1
26	2	1	3	0	1
27	3	1	2	0	1
28	3	1	2	0	1
29	3	1	2	0	1
30	2	1	3	0	1
31	3	1	2	0	1
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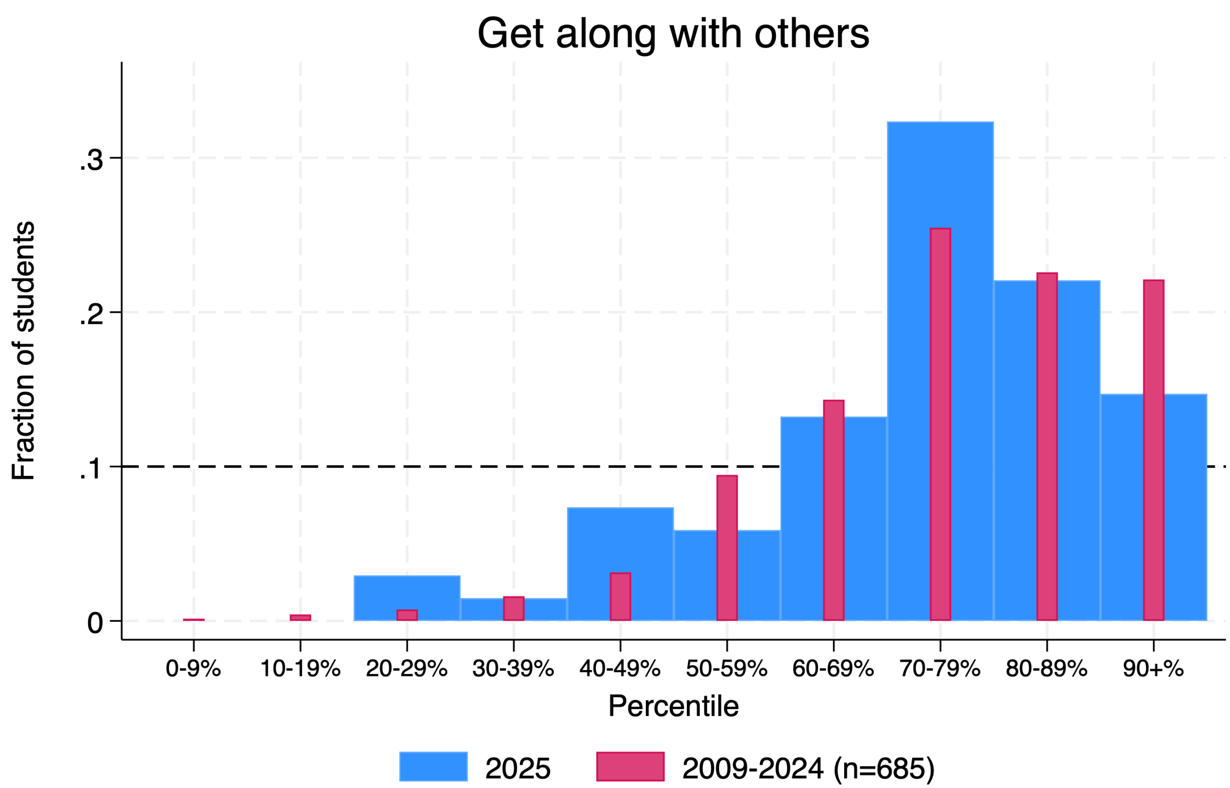
42	1	1	3	1	0
43	2	1	3	0	1
44	2	1	3	0	1
45	3	1	2	0	1
46	1	1	2	1	0
47	1	1	3	1	0
48	1	1	3	1	0
49	2	1	3	0	1
50	3	1	2	0	1

Bonus Example I

Question:

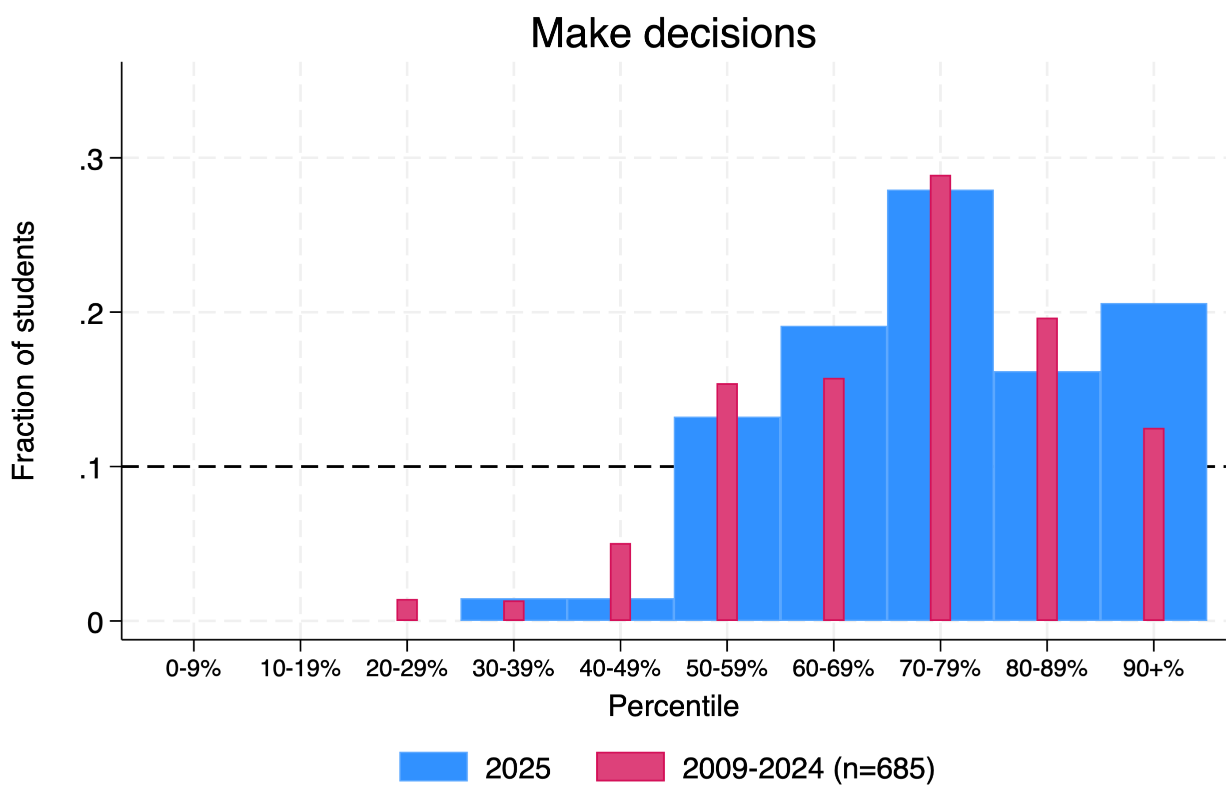
- Compared to the other students in this class, how do you think you rate in ability to <u>get along with others</u>?



Bonus Example II

Question:

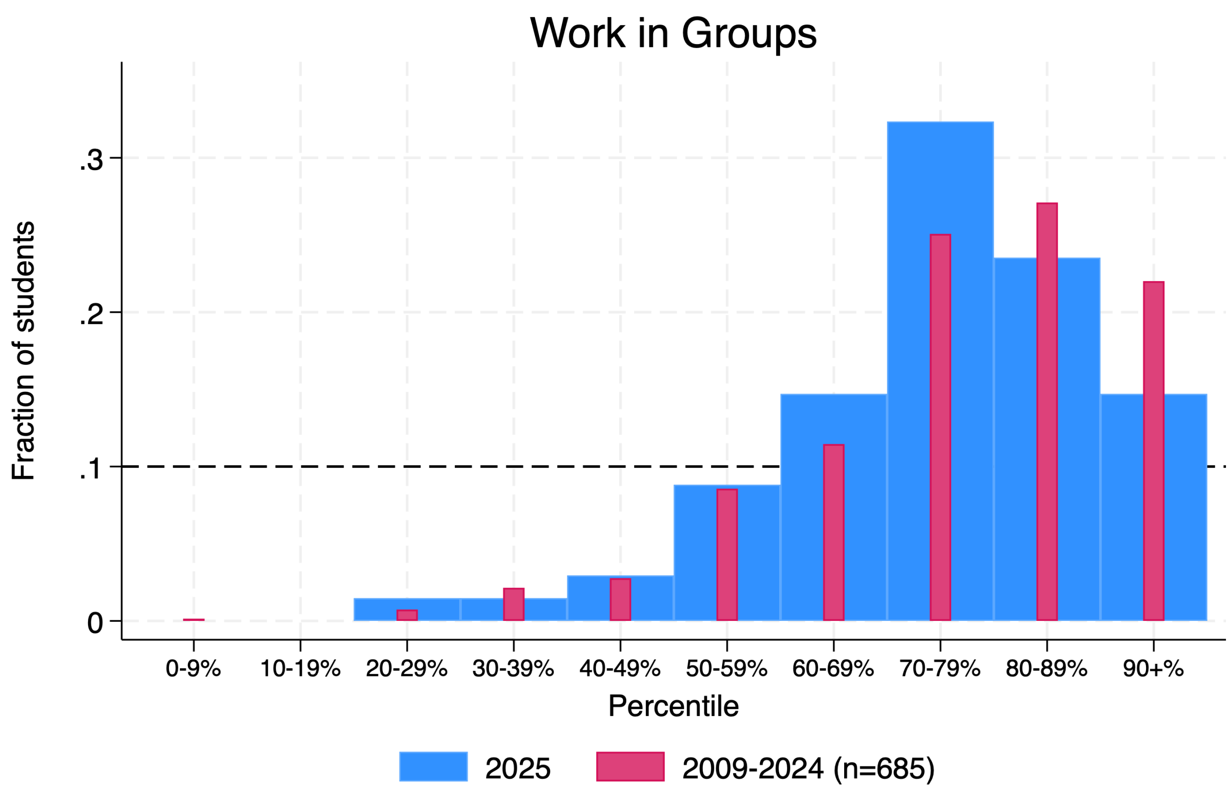
- Compared to the other students in this class, how do you think you rate in ability to make decisions?



Bonus Example III

Question:

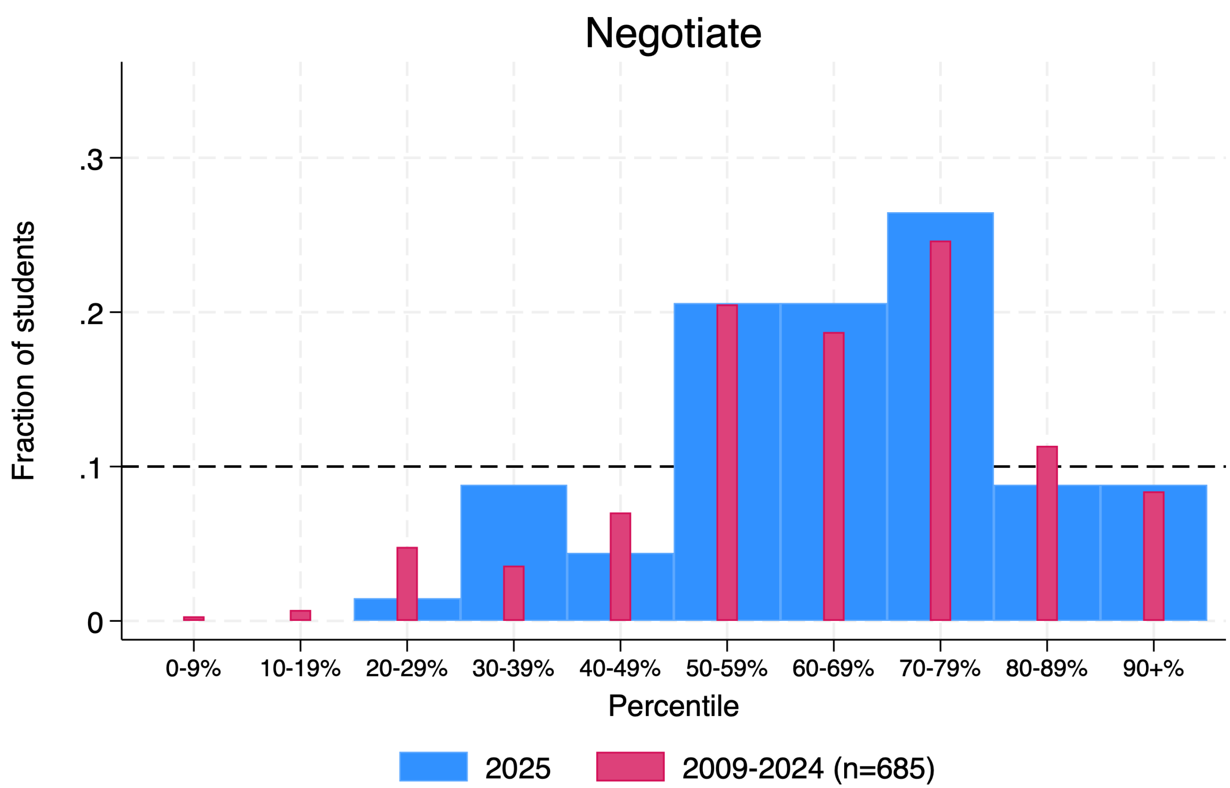
- Compared to the other students in this class, how do you think you rate in ability to <u>work in groups</u>?



Bonus Example IV

Question:

- Compared to the other students in this class, how do you think you rate in ability to <u>negotiate</u>?



Prescriptions

- Example 1 (Professor)
  - don’t wimp out!
- Example 2 (Geometry)
  - use normative rule
- Example 3 (Müller-Lyer)
  - use normative rule (or outside reference/instrument [cf. pilots])
- Example 4 (Framing of decisions)
  - consider alternative descriptions
- Example 5 (Vacation)
  - don’t let uncertainty disable decision making
- Example 6 (Monty Hall Problem)
  - “when in doubt, simulate!”
- Bonus Example (Positive Illusion)
  - seek out feedback!

Why Study Decision Making?

- Improve your own decision making
  - Individual: How can you make better and faster decisions?
  - Organizational: How can you get your organization (and/or bosses, peers, subordinates) to make better and faster decisions?

- Understand how others make decisions
  - Marketing: How do people decide what to buy?
  - Finance: How do people/firms decide how to invest? What does that imply about prices?
  - Organizations: Why do organizations make decisions the way they do?
  - Venture Capital: How do entrepreneurs make decisions?

Why Study Decision Making?

- Improve decision environments
  - Create better *choice architecture* to minimize decision biases, simplify choices, etc.

Course Outline

Monday, November 3	Introduction to Decision Making
Tuesday, November 4	Judgment under Uncertainty I (Heuristics and biases)
Wednesday, November 5	Judgment under Uncertainty II (The difficulty of learning)
Friday, November 7	Risky Decision Making
Saturday, November 8	Nudges/Improving Decision Processes
January 5	Final (More later)

Course Requirements

- Class Participation (25%)
  - Quality of participation
  - Cold calls
- Final (35% to 50%)
  - Sample Questions in Classes 2 and 4
- Prep Notes (10%)
  - Group case web assignments (Class 1, 3, 4, and 5)
  - Individual assignments (Classes 2 and 5)
- Debrief Notes (15% to 30%)
  - Group assignments (Class 1, 2, 3, 4, and 5)

Daily Debrief Form

- Goal
  - Help you connect class concepts with your professional lives
- Organization
  - **Identify** a real-world example that captures a concept discussed in class
  - Use this concept to better **understand** that situation
  - Come up with practical ways to **improve** the decision-making in that situation
- More details on Canvas
- Submit
  - Through Canvas by 9:00 AM the next day

Probability Assessment Questionnaire

Class 2 Prep Note (No Group Assignment)

- Access web survey
  - Link on Canvas (Assignment 2)

- Complete the survey
- Due 9:00 PM tonight!

Probability Assessment

"Uncertain" Quantity: Market Capitalization of Bank of China, November 3, 2025 (billion HK$)

中國銀行(香港)
BANK OF CHINA (HONG KONG)

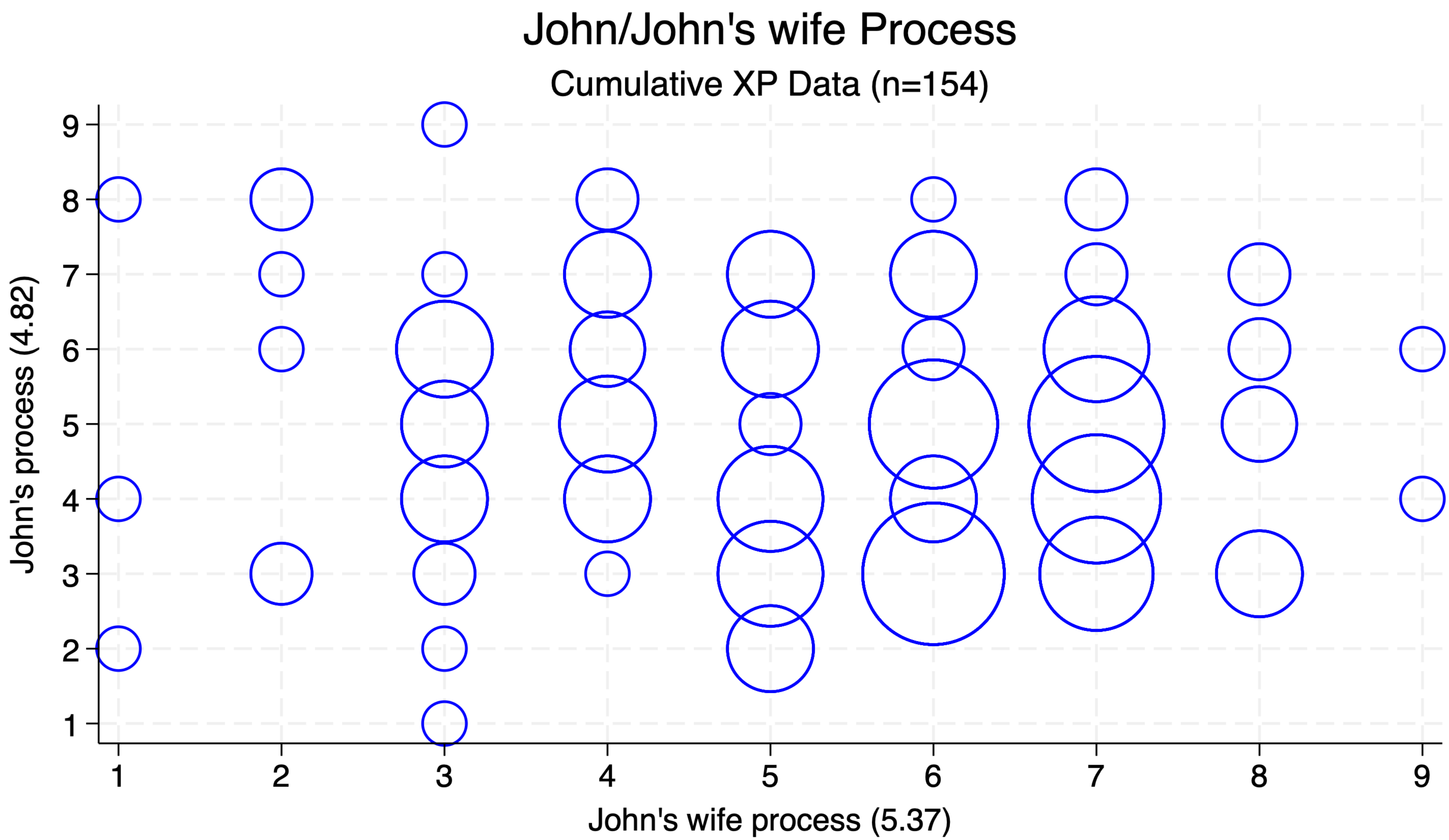
.05	.50	.95

Answer:

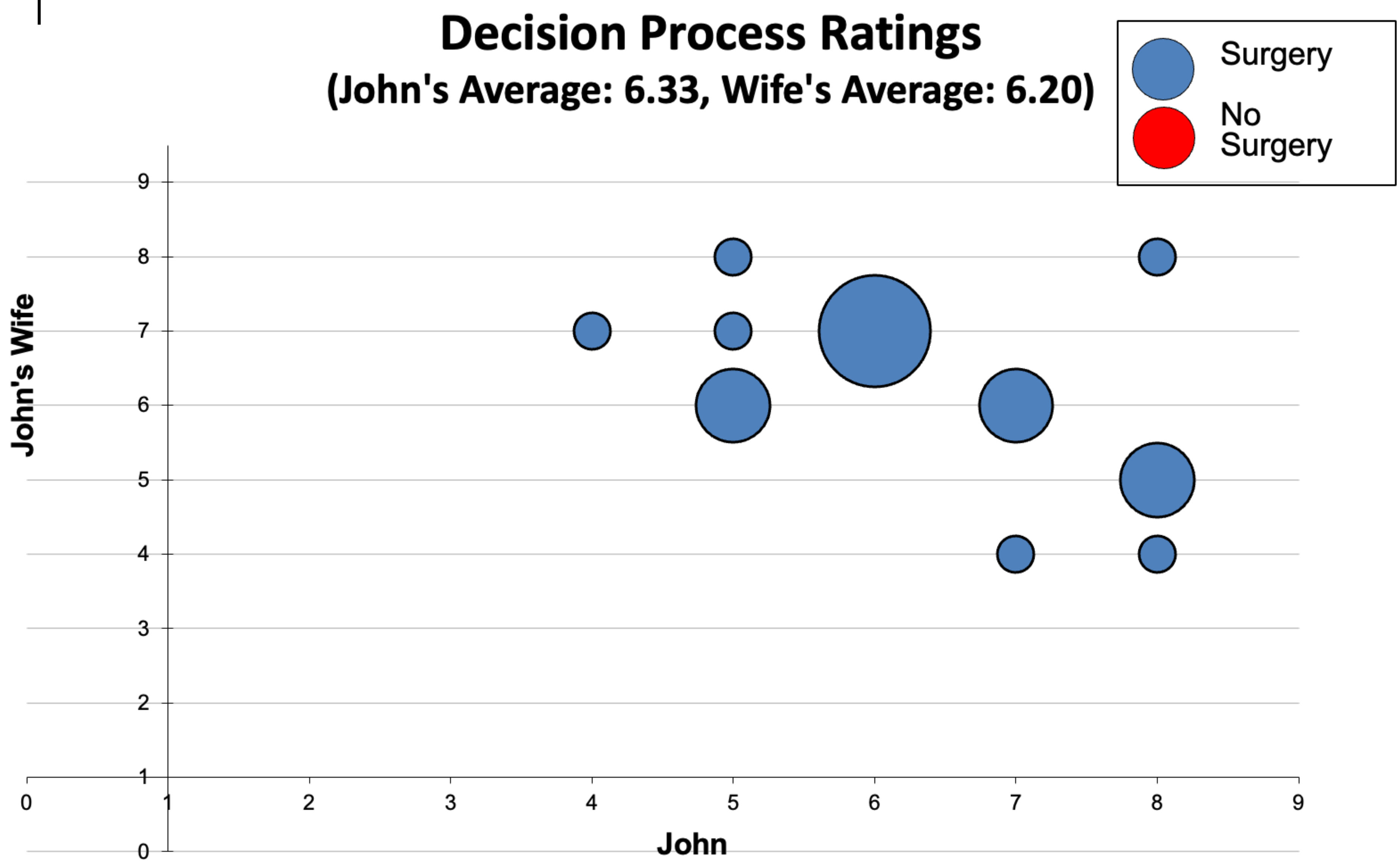
John Brown

"The core predicament of medicine – the thing that makes being a patient so wrenching, being a doctor so difficult, and being a part of a society that pays the bills they run up so vexing – is uncertainty … Medicine's ground state is uncertainty. And wisdom – for both patients and doctors – is defined by how one copes with it."

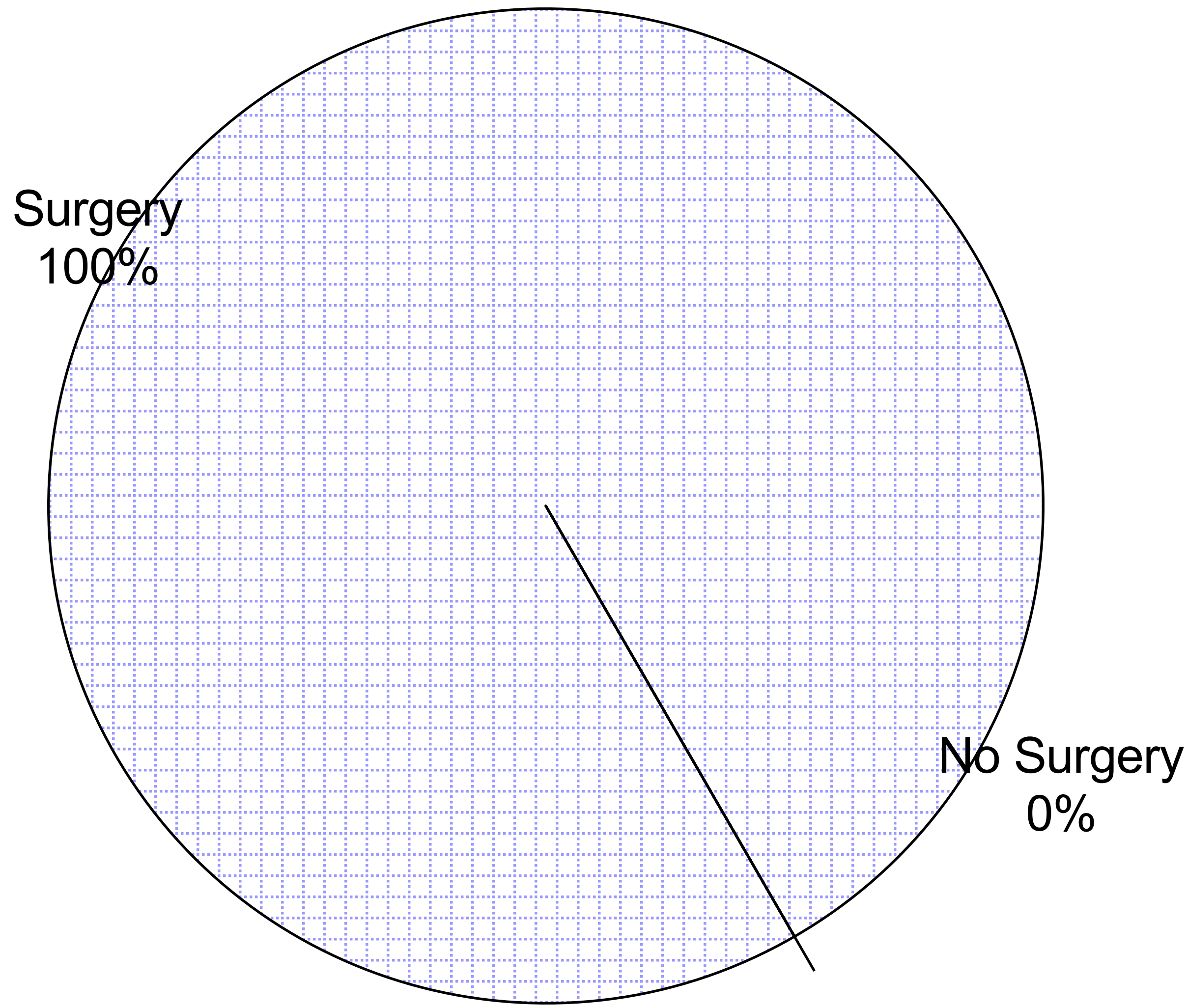
- Atul Gawande (2002), *Complications* ("The case of the red leg")

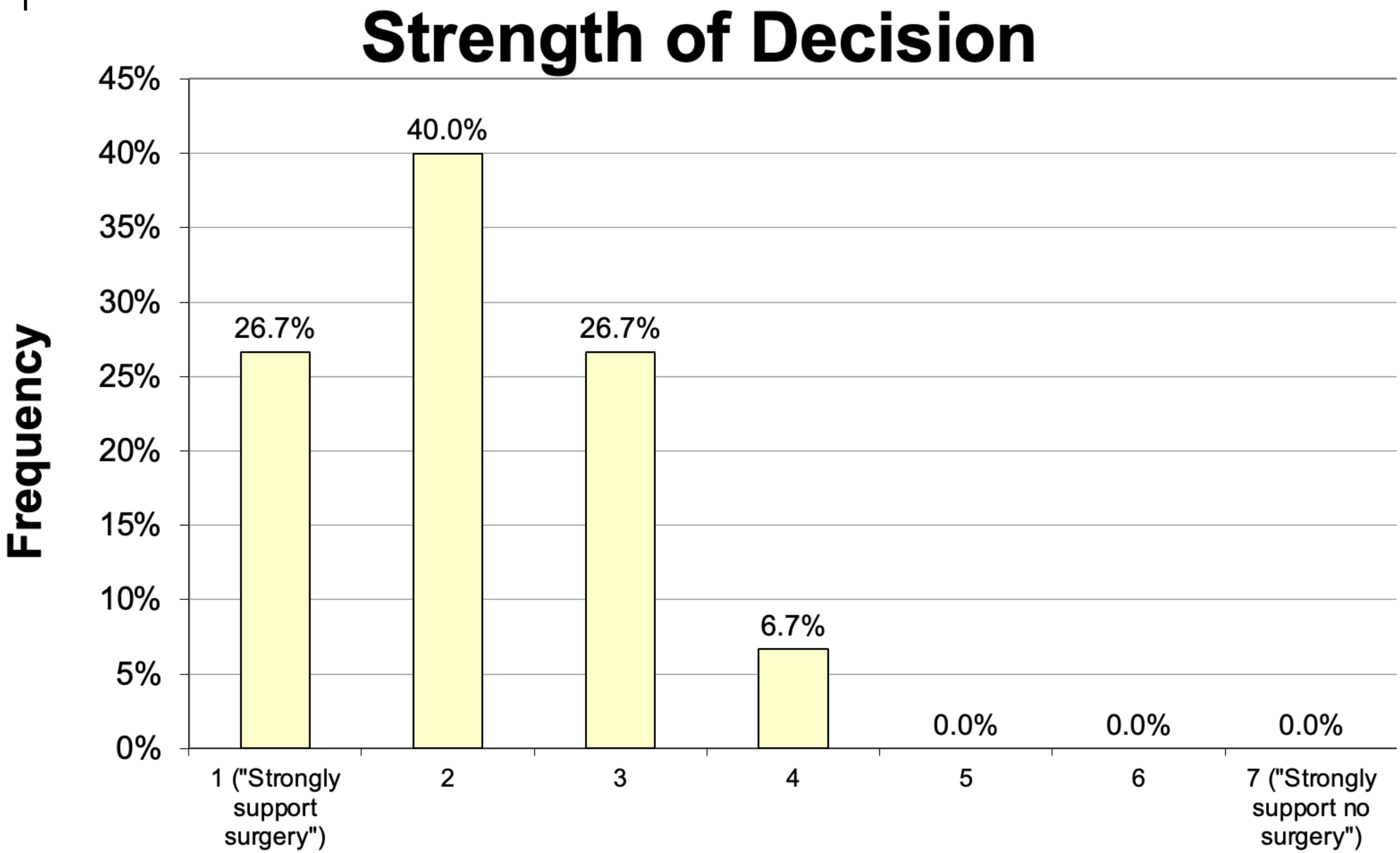


John/John's wife Process
Cumulative XP Data (n=154)
John's process (4.82)
John's wife process (5.37)
1
2
3
4
5
6
7
8
9

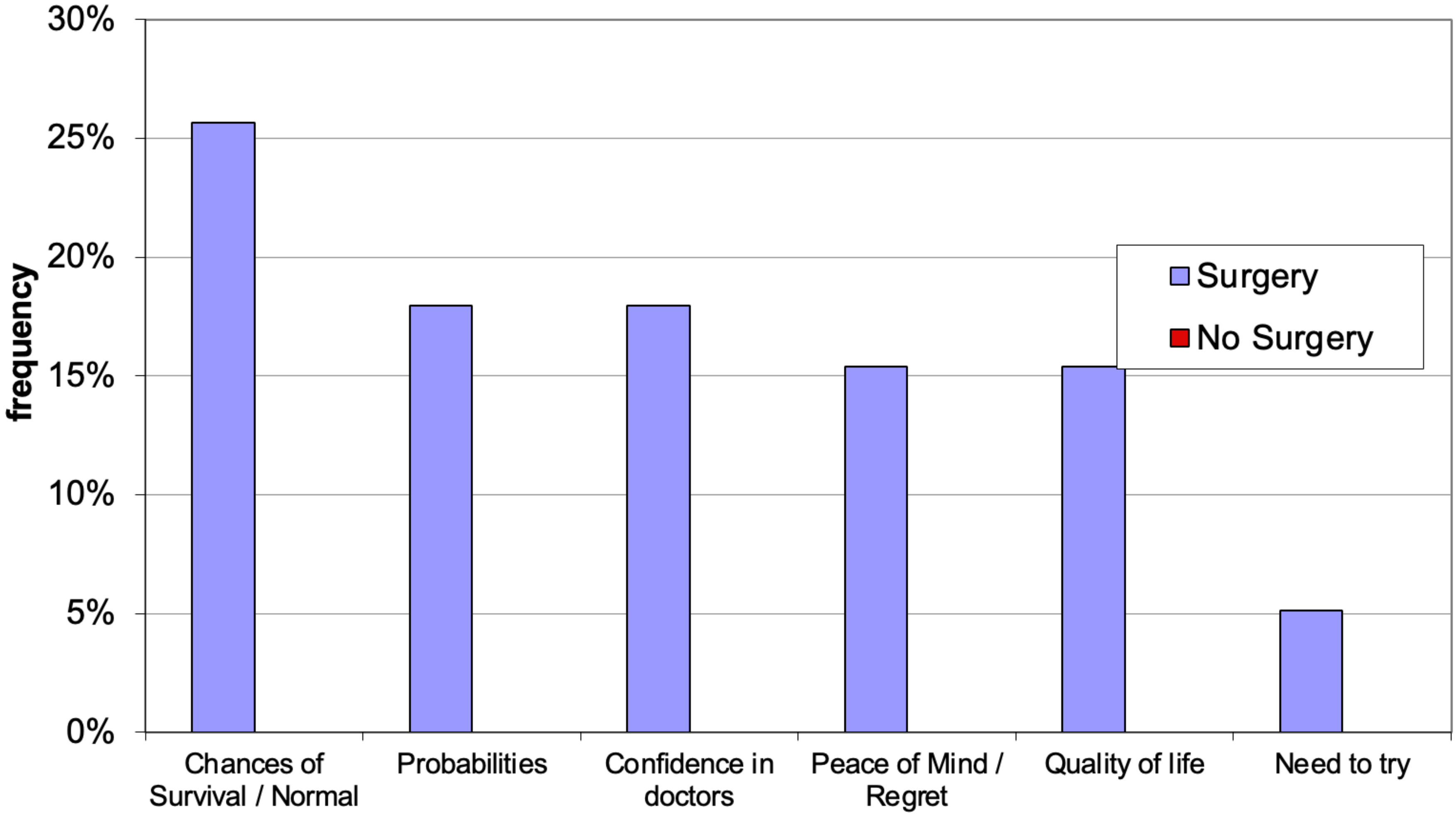

Decision Process Ratings
(John's Average: 6.33, Wife's Average: 6.20)
Surgery
No Surgery
John's Wife
John
0
1
2
3
4
5
6
7
8
9

Surgery Decision

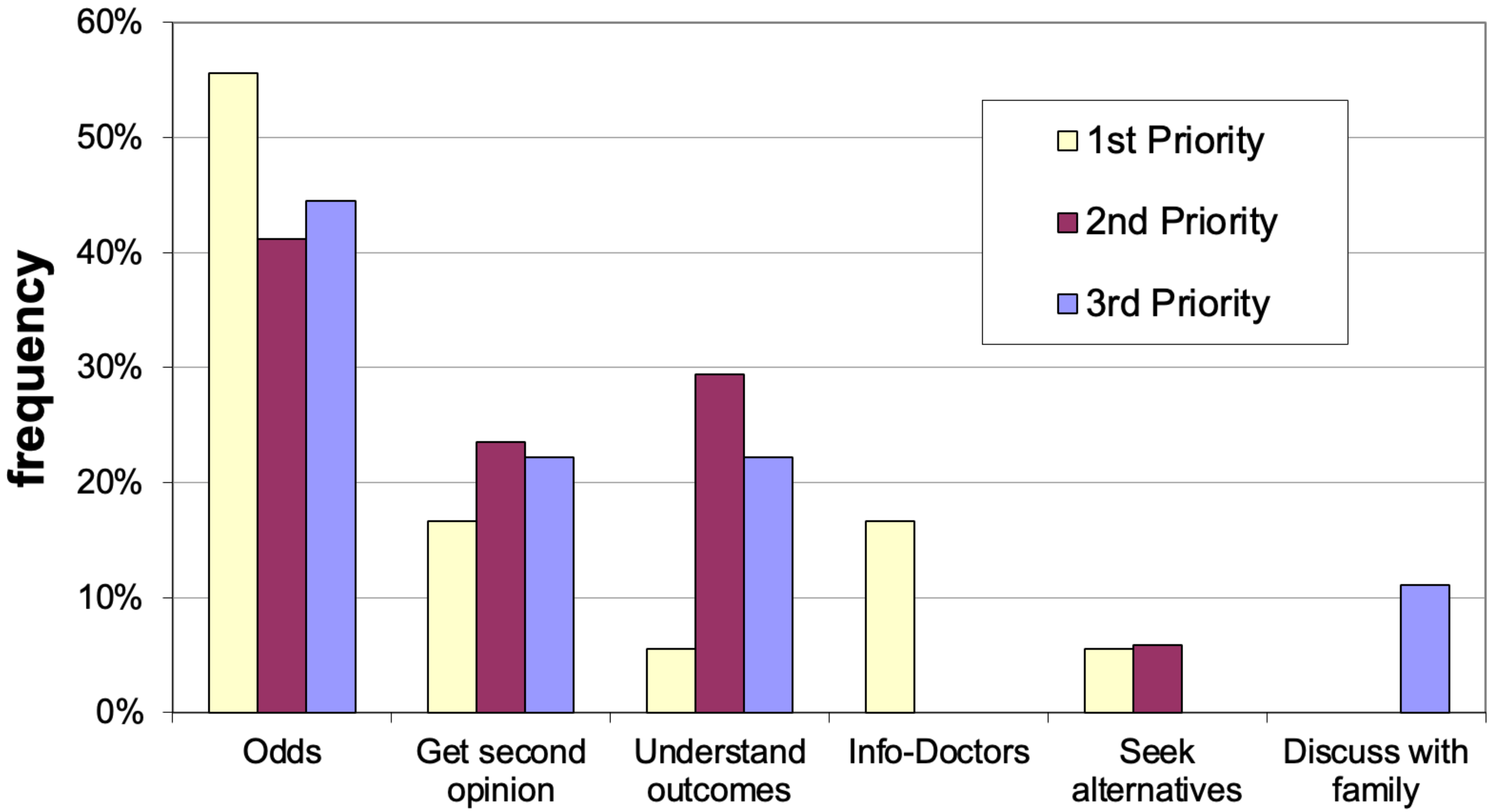


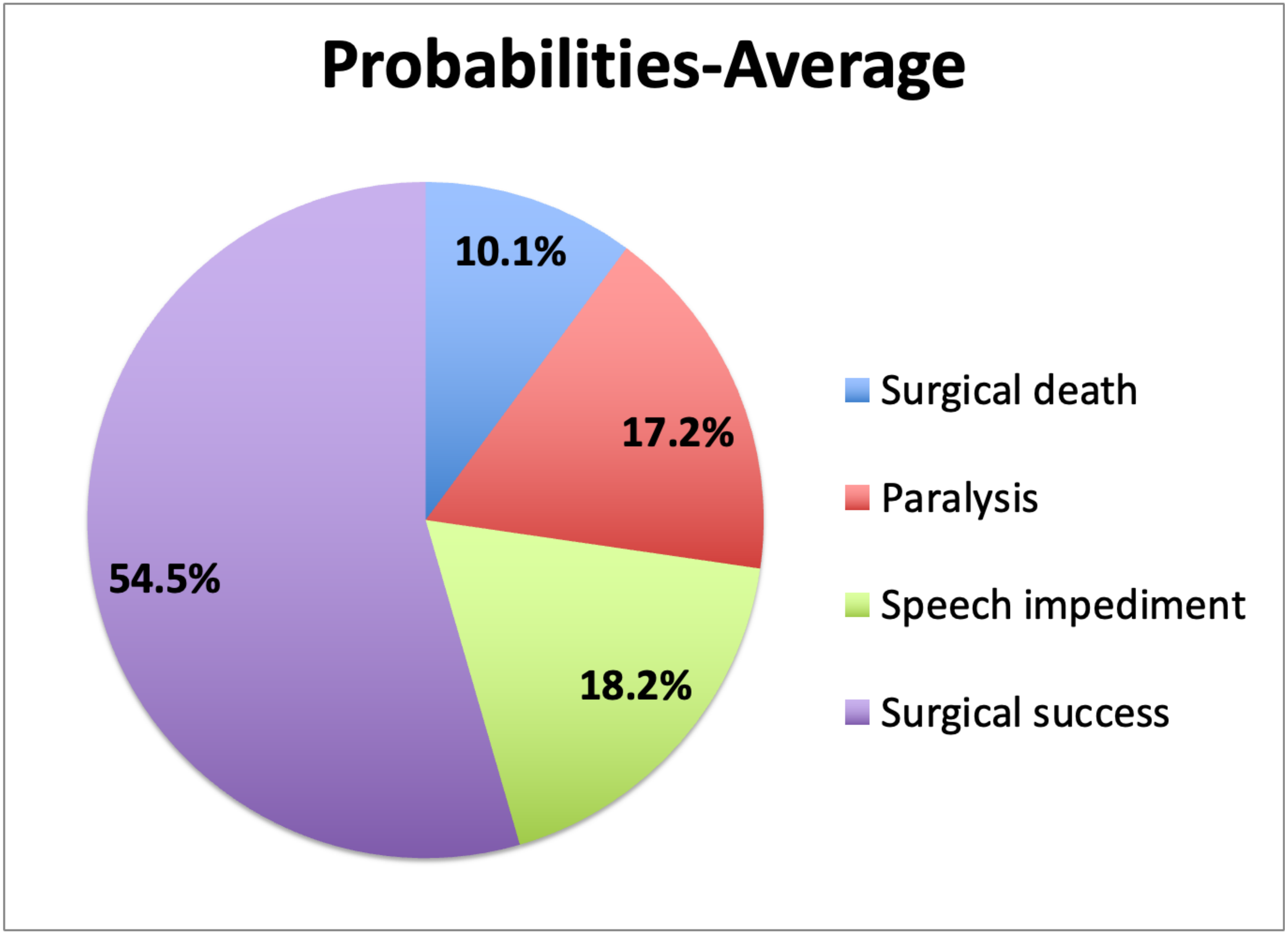

Strength of Decision
Frequency
45%
40%
35%
30%
25%
20%
15%
10%
5%
0%
26.7%
40.0%
26.7%
6.7%
0.0%
0.0%
0.0%
1 ("Strongly support surgery")
2
3
4
5
6
7 ("Strongly support no surgery")

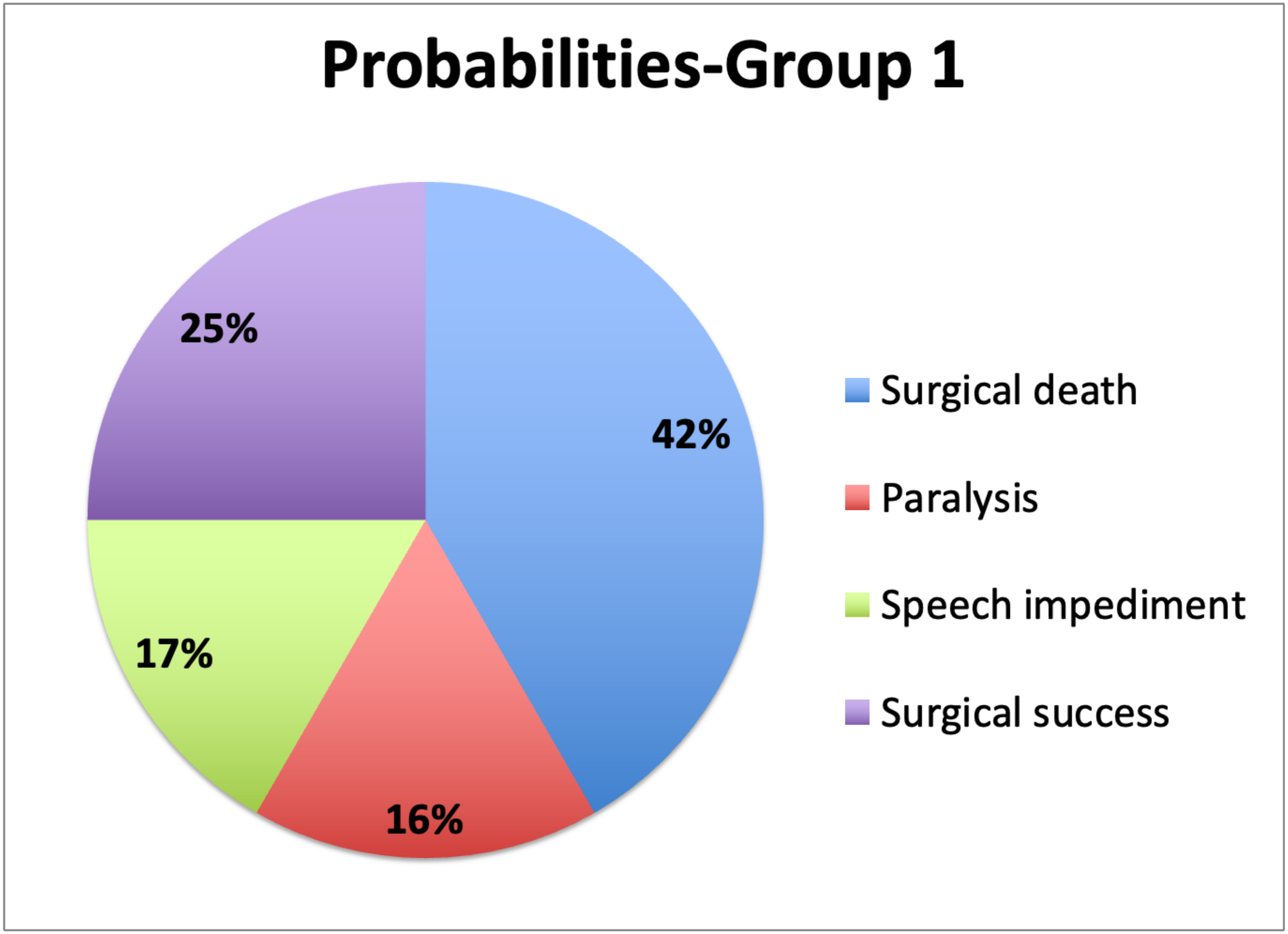
Reasons for Decision

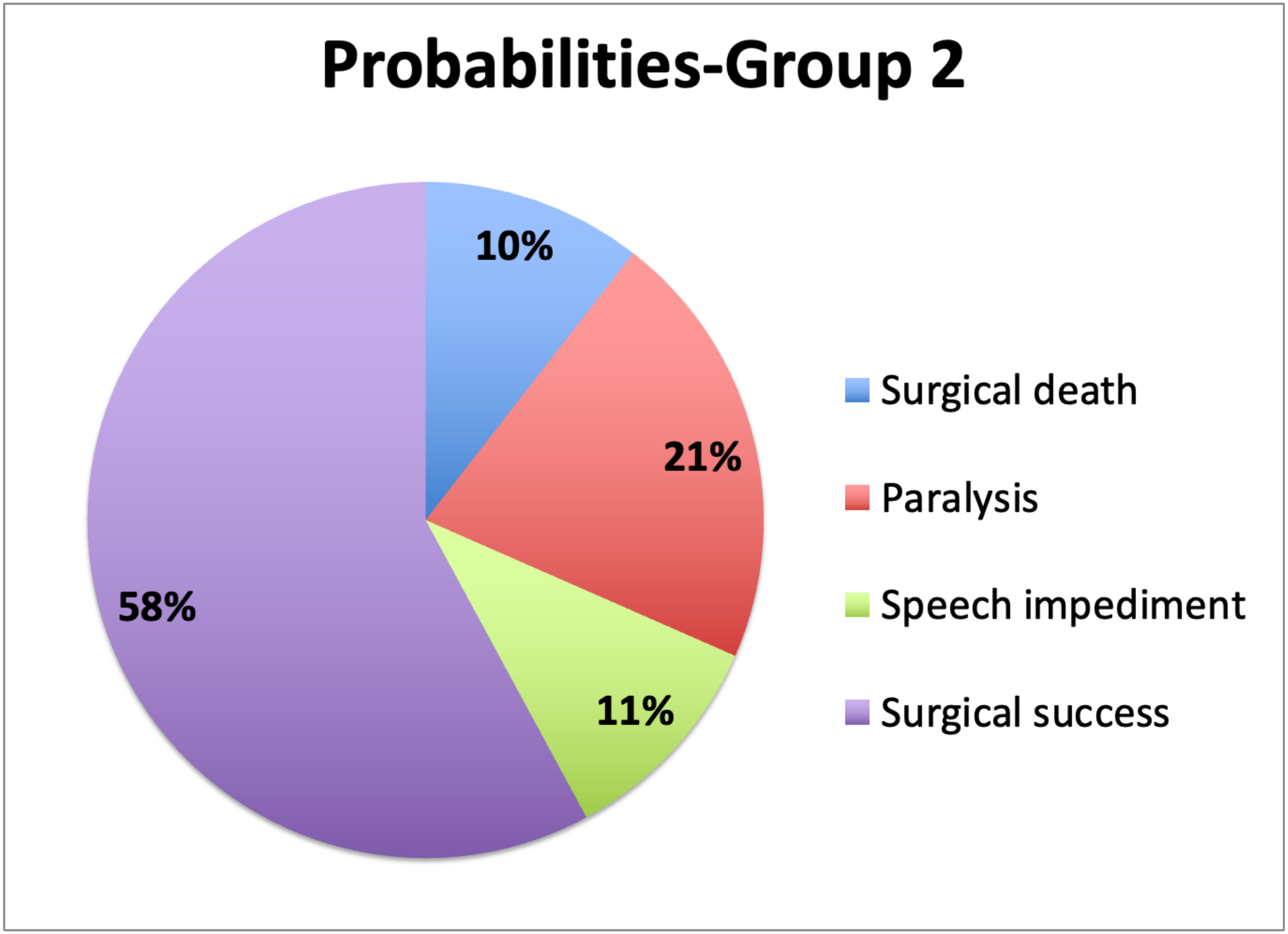


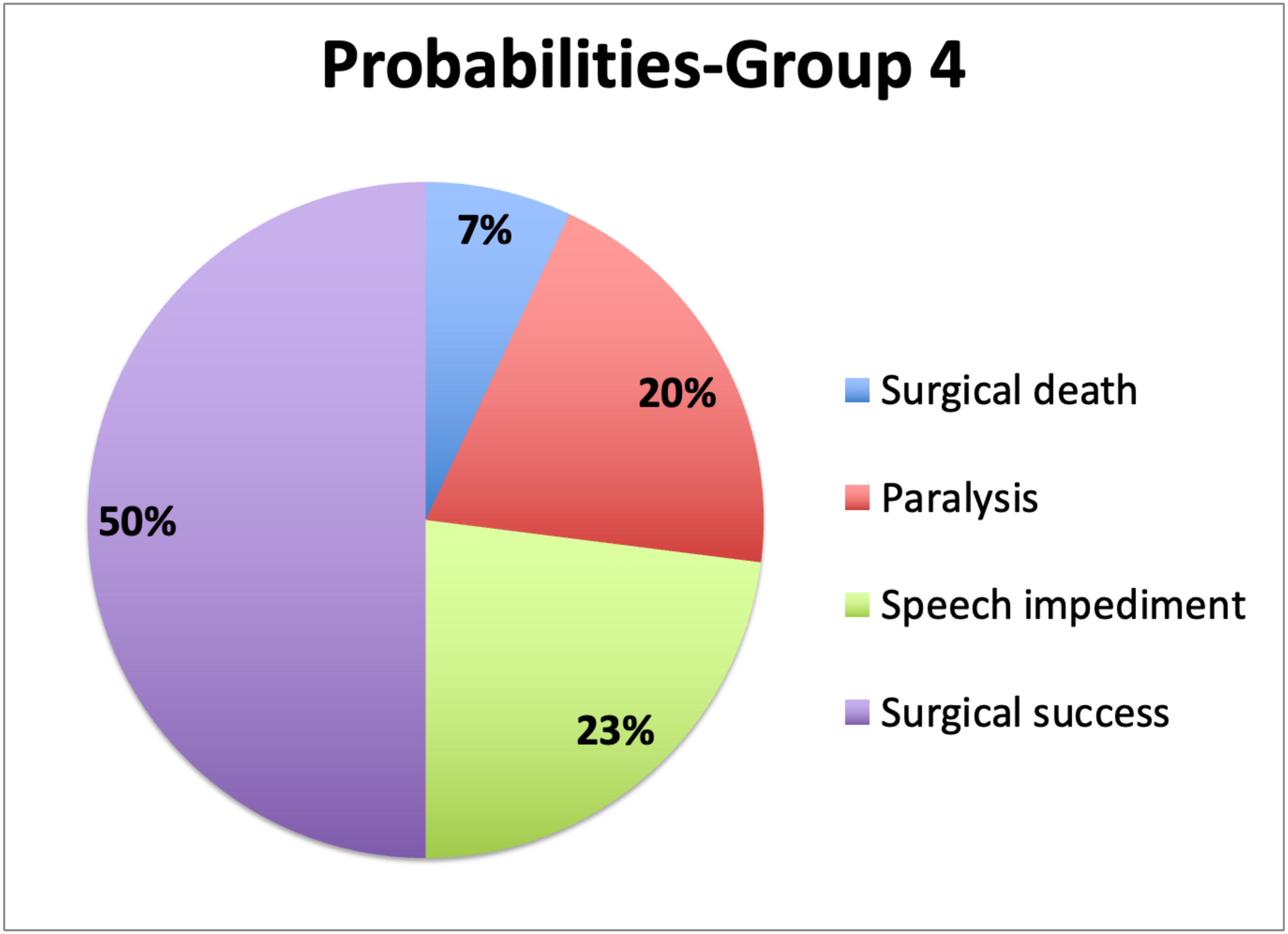
Use of Extra Time

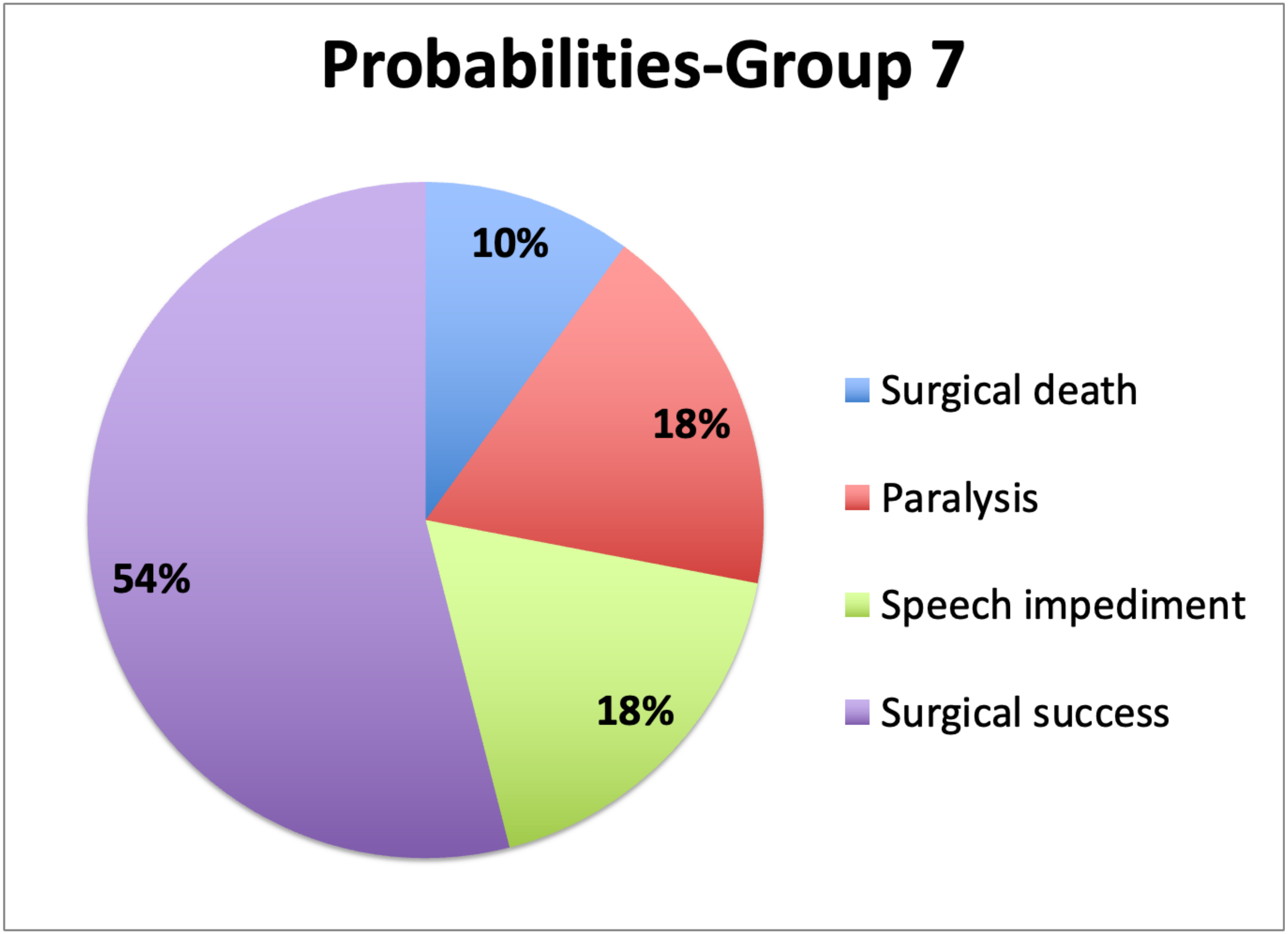


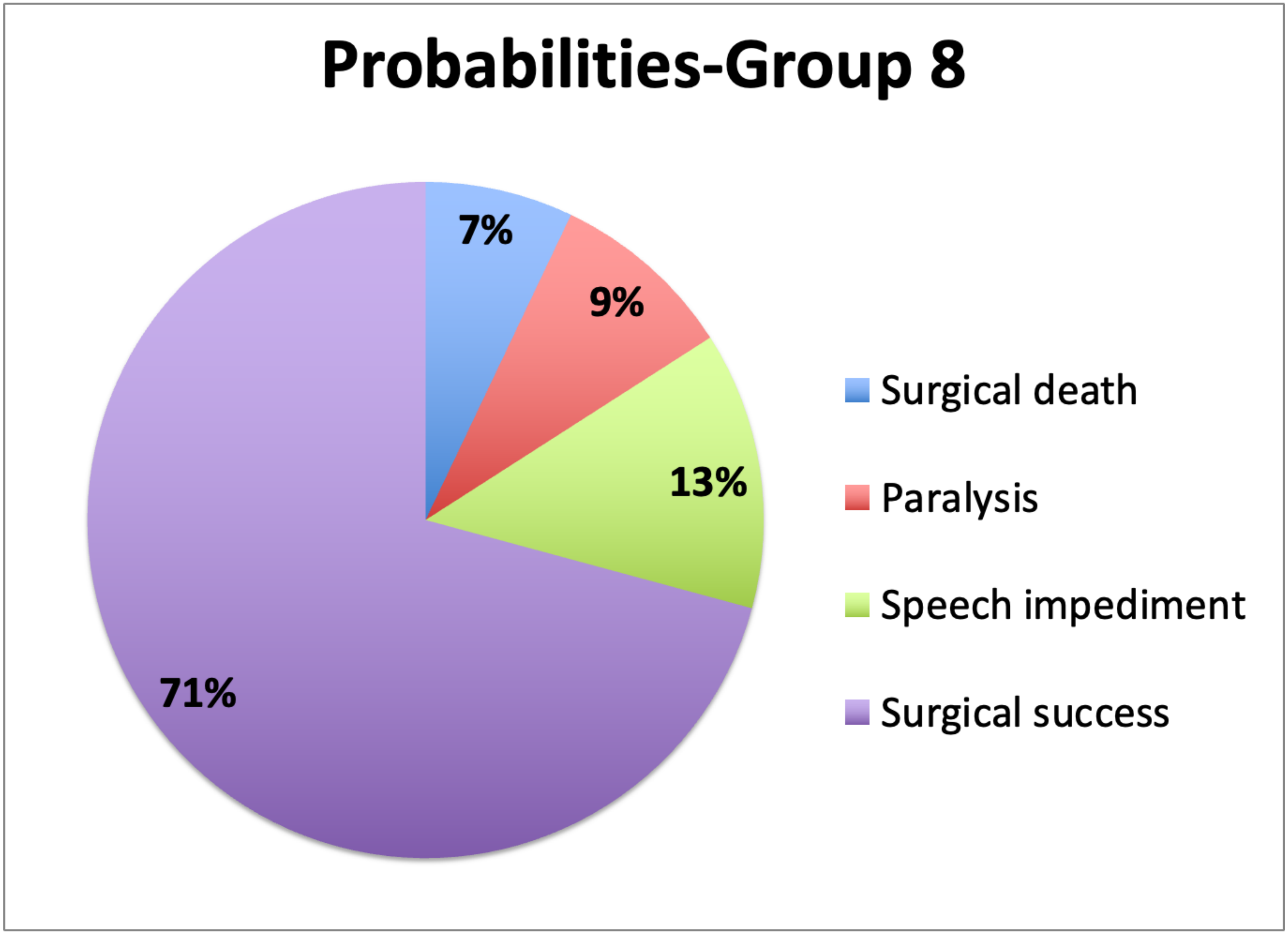

Probabilities-Average
10.1%
17.2%
18.2%
54.5%
Surgical death
Paralysis
Speech impediment
Surgical success

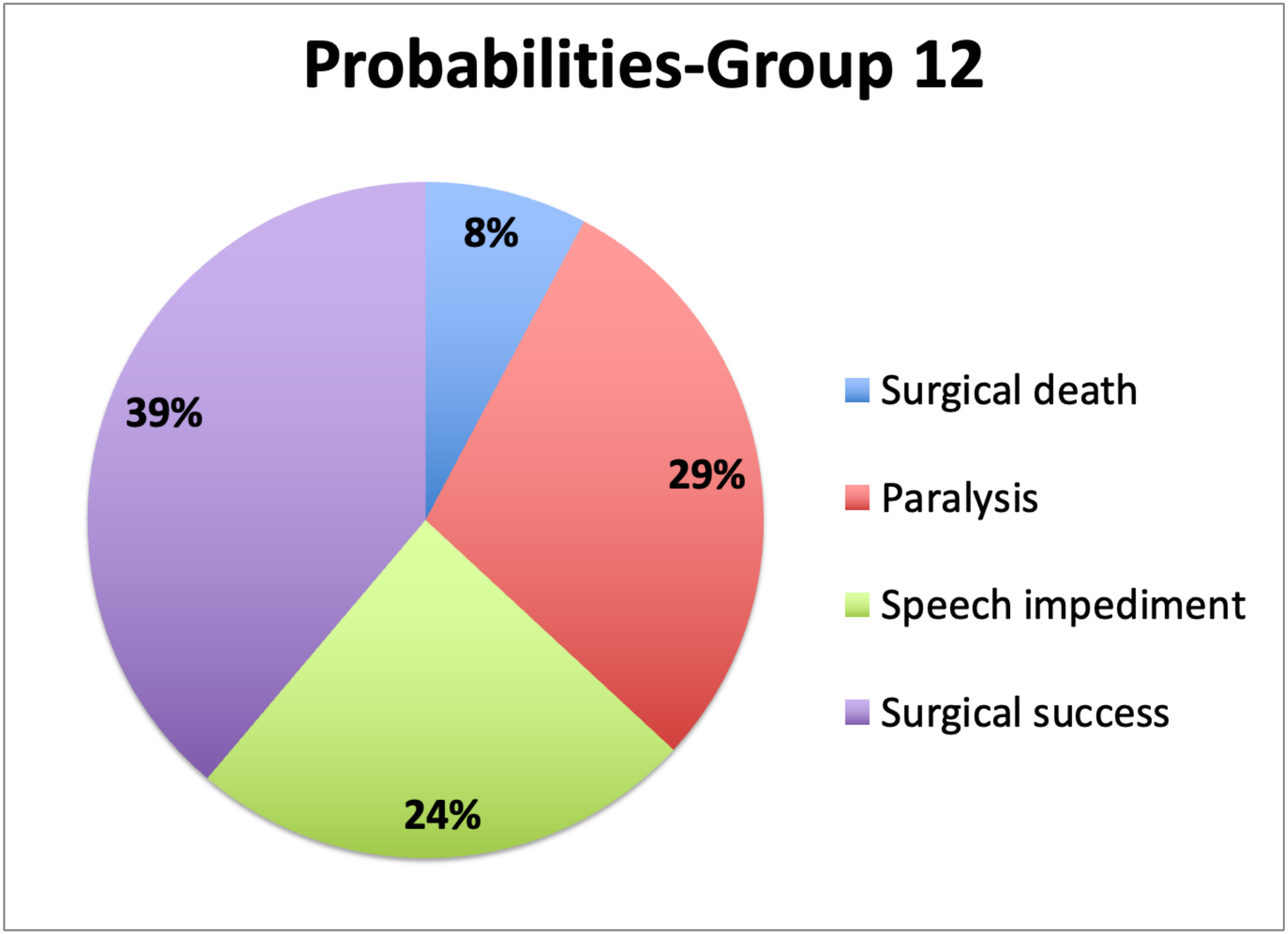

Probabilities-Group 1
25%
42%
17%
16%
Surgical death
Paralysis
Speech impediment
Surgical success

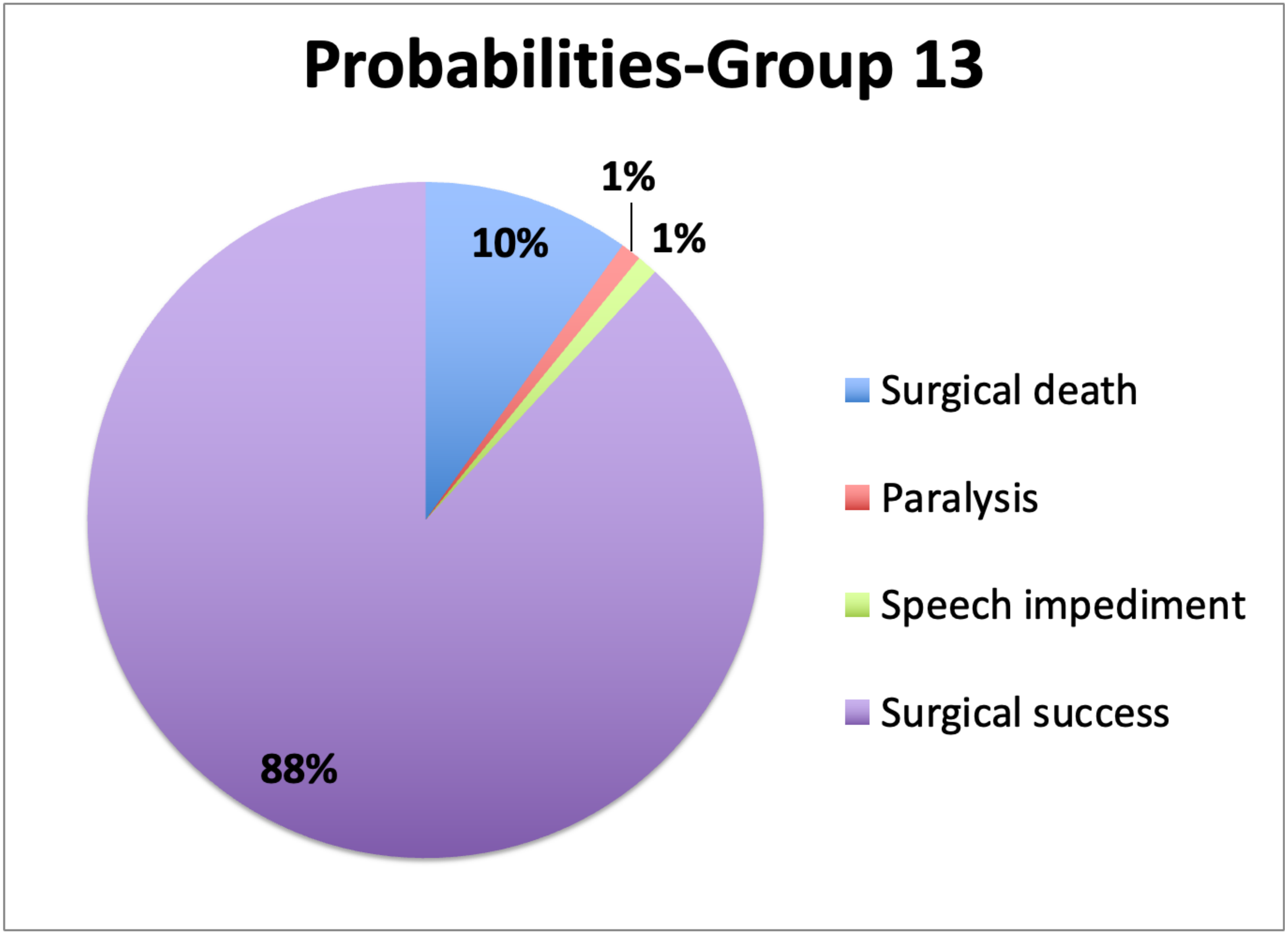

Probabilities-Group 2
10%
21%
11%
58%
Surgical death
Paralysis
Speech impediment
Surgical success

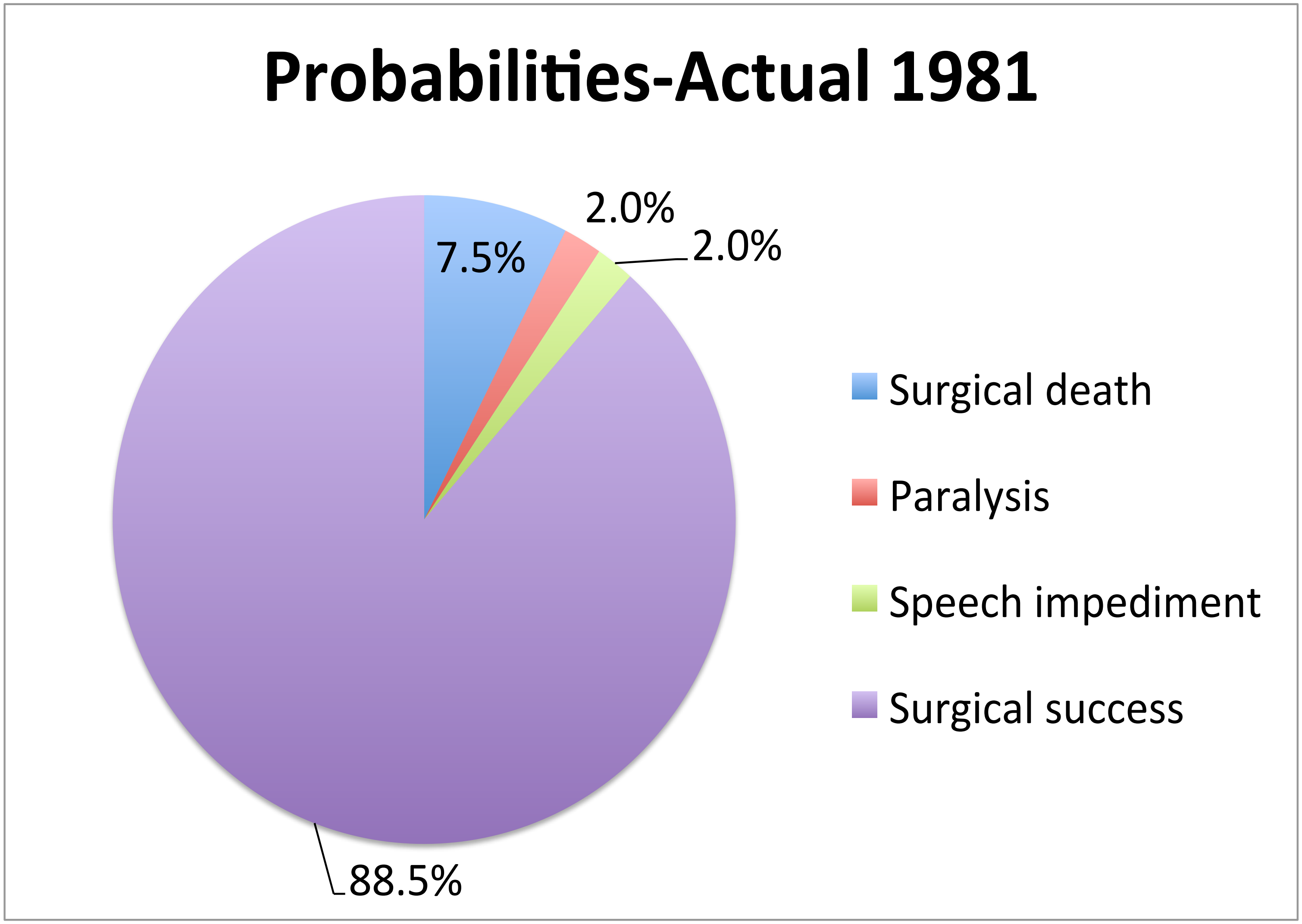

Probabilities-Group 4
7%
20%
50%
23%
Surgical death
Paralysis
Speech impediment
Surgical success

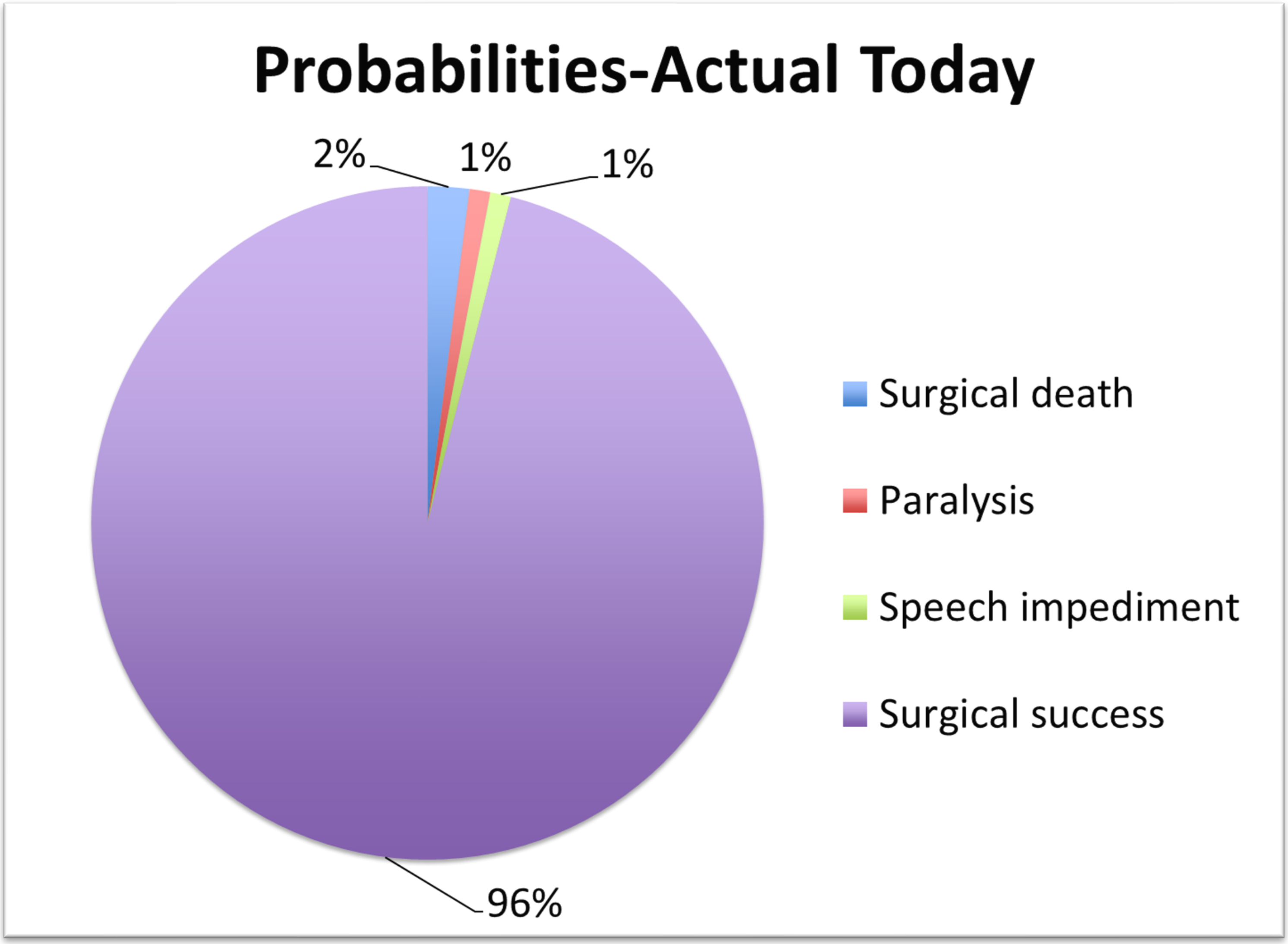

Probabilities-Group 7
10%
18%
18%
54%
Surgical death
Paralysis
Speech impediment
Surgical success


Probabilities-Group 8
7%
9%
13%
71%
Surgical death
Paralysis
Speech impediment
Surgical success


Probabilities-Group 12
8%
29%
24%
39%
Surgical death
Paralysis
Speech impediment
Surgical success


Probabilities-Group 13
10%
1%
1%
88%
Surgical death
Paralysis
Speech impediment
Surgical success


Probabilities-Actual 1981
7.5%
2.0%
2.0%
88.5%
Surgical death
Paralysis
Speech impediment
Surgical success


Probabilities-Actual Today
2%
1%
1%
96%
Surgical death
Paralysis
Speech impediment
Surgical success

Wu's Clues: Part 1

Clue 1: What's intuitive isn't necessarily what's best.

Clue 2: Don't confuse a good decision with a good outcome.

Clue 3: Analysis helps, but don't let the hard drive out the soft.

Clue 4: Don't relinquish an important decision to someone just because that person is an expert.

Clue 5: Smart decisions can be better and faster.